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E³UDRES²

Multimodal collection of biometric data related to stress in XR interventions

Development of a library for data collection and pilot study

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Abstract

Multimodal biometric data collected during extended reality (XR) interventions can provide a more comprehensive understanding of participants' physiological and psychological responses. However, collecting such data often requires specialized equipment and trained personnel, increasing costs and limiting the amount of data that can be collected.

This thesis presents the design, implementation, and evaluation of a software library for collecting electrocardiogram (ECG) data from a Polar H10 heart rate monitor, head-movement data from a Meta Quest 3 headset, and voice recordings. The library supports participant-operated data collection in supervised on-site and unsupervised at-home settings.

A prototype was integrated into a VR application that collected heart rate, ECG, head-movement, and voice data while participants completed a Stroop Room task. It was evaluated on-site with 14 participants using the System Usability Scale (SUS) and at home with two participants over one week. Usability, data completeness, and the suitability of the collected biomarkers for stress research were assessed.

The on-site evaluation showed high usability (mean SUS score: 85), and all participants reported increased stress after the task. The take-home evaluation demonstrated the feasibility of independent multimodal data collection, although raw ECG data were incomplete in some sessions. Heart-rate variability (HRV) responses varied substantially between participants, while head-movement data enabled above-chance classification of task phases.

Overall, the library and VR application provide a usable and feasible foundation for supervised and at-home multimodal data collection. However, improved quality-control mechanisms and larger studies are needed to establish physiological and movement features as reliable stress biomarkers.

Kurzfassung

Multimodale biometrische Daten, die während Interventionen in erweiterter Realität (Extended Reality, XR) erhoben werden, können ein umfassenderes Verständnis der physiologischen und psychologischen Reaktionen von Teilnehmenden ermöglichen. Die Erhebung solcher Daten erfordert jedoch häufig spezielle Geräte und geschultes Personal, wodurch die Kosten steigen und der Umfang der Datenerhebung begrenzt wird.

Diese Masterarbeit beschreibt den Entwurf, die Implementierung und die Evaluierung einer Softwarebibliothek zur Erfassung von Elektrokardiogramm-Daten (EKG) mit einem Polar-H10-Herzfrequenzsensor, von Kopfbewegungsdaten mit einem Meta-Quest-3-Headset sowie von Sprachaufzeichnungen. Die Bibliothek ermöglicht eine von den Teilnehmenden selbstständig durchgeführte Datenerhebung sowohl in betreuten Vor-Ort-Settings als auch in unbeaufsichtigten häuslichen Settings.

Ein Prototyp wurde in eine Virtual-Reality-Anwendung (VR) integriert, die Herzfrequenz-, EKG-, Kopfbewegungs- und Sprachdaten erfasste, während die Teilnehmenden eine Stroop-Room-Aufgabe absolvierten. Der Prototyp wurde vor Ort mit 14 Teilnehmenden mithilfe der System Usability Scale (SUS) sowie über einen Zeitraum von einer Woche zu Hause mit zwei Teilnehmenden evaluiert. Dabei wurden die Benutzerfreundlichkeit, die Vollständigkeit der Daten und die Eignung der erhobenen Biomarker für die Stressforschung untersucht.

Die Vor-Ort-Evaluierung ergab eine hohe Benutzerfreundlichkeit mit einem durchschnittlichen SUS-Wert von 85. Zudem berichteten alle Teilnehmenden nach der Aufgabe von einem erhöhten Stressempfinden. Die Evaluierung im häuslichen Umfeld zeigte, dass eine selbstständige multimodale Datenerhebung grundsätzlich durchführbar ist, wenngleich die EKG-Rohdaten in einigen Sitzungen unvollständig waren. Die Reaktionen der Herzfrequenzvariabilität (HRV) unterschieden sich deutlich zwischen den Teilnehmenden, während die Kopfbewegungsdaten eine Klassifikation der Aufgabenphasen mit einer Genauigkeit oberhalb des Zufallsniveaus ermöglichten.

Insgesamt bieten die Softwarebibliothek und die VR-Anwendung eine benutzerfreundliche und praktikable Grundlage für die betreute multimodale Datenerhebung sowie für deren Durchführung im häuslichen Umfeld. Es sind jedoch verbesserte Mechanismen zur Qualitätskontrolle und umfangreichere Studien erforderlich, um physiologische Merkmale und

Bewegungsmerkmale als zuverlässige Stressbiomarker zu etablieren.

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1 Introduction

Virtual Reality (VR) is used in an increasing number of applications in the medical field. Applications include training, rehabilitation, psychotherapy and surgery (Riener & Harders, 2012). One particularly interesting and active area of research is the use of VR in stress management and resilience training. Haeussel et al. (2026) and Meshkat et al. (2024) reviewed multiple studies in that area.

A common practice in stress-related VR interventions is, besides administering standardized questionnaires, to record physiological biomarkers associated with stress. These biomarkers provide researchers with information about participants' autonomic nervous system (ANS) responses to the VR interventions.

Carrying out the VR interventions and recording data often require substantial time and effort. Participants need to come to a lab, a technician needs to set up the recording apparatus and sometimes schedule multiple sessions. This thesis introduces a system intended to enable participants to record head movements, voice, and heart-rate data independently at home using a commercially available VR headset and heart-rate monitoring device.

1.1 Research Goals

This thesis pursues two primary research goals. First, it seeks to identify and define suitable biometric parameters obtained from a Polar H10 heart rate monitor (HRM) and a Meta Quest 3 VR headset for the purpose of establishing a physiological stress baseline. These parameters may include cardiovascular indicators such as heart rate (HR) and heart rate variability (HRV), as well as additional physiological measures captured by the VR system. Second, the thesis aims to develop and implement a structured process for the longitudinal recording and storage of stress baseline data over a longer period. This process is intended to ensure consistent data collection and secure data management, thereby enabling subsequent analysis and further scientific processing by the research staff following the intervention.

1.2 Research Questions

To pursue the goals stated in section 1.1, this thesis intends to answer the following research questions:

- To what extent can a standalone VR application and reusable data-collection library support the collection and storage of multimodal data in supervised and home-based settings?
 - How do HRV features vary across the phases of the VR protocol, and how are they associated with self-reported stress?
 - Which movement parameters tracked by the Meta Quest 3 can be recorded and stored for the analysis of stress-related responses, and to what extent can they distinguish between the baseline and cooldown phases of the VR protocol?
 - How complete and usable are repeated multimodal recordings collected independently at home?

1.3 Motivation

Despite promising developments in XR stress intervention research, an important practical question remains as to how such interventions can be implemented as autonomous systems for regular use outside the laboratory. Workplace-based VR interventions are valuable, but their use can be limited by staffing constraints, workload, and the difficulty of taking protected breaks during shifts, as noted in nurse-focused research Nijland et al. (2021). Home-based delivery offers a complementary model in which participants can engage with stress interventions more flexibly and repeatedly in their own environment, and early findings suggest that such use is feasible and acceptable (Pallavicini et al., 2022). Building on this development, the present thesis focuses on how an XR stress recording system can be designed for self-guided use at home. Its relevance lies in extending immersive stress recording beyond supervised and workplace-bound settings and in exploring how multimodal biometric data can be captured by such a system. In this way, the work contributes to the development of XR interventions that are not only immersive, but also suitable for independent, real-world stress interventions.

1.4 Thesis structure

This thesis begins by introducing the necessary concepts and background information in Chapter 2, which provide the foundation for understanding the remainder of the work. Chapter 3 presents the methodology and results of a systematic literature review that aims to describe the current state of the art in measuring stress-related biomarkers in VR interventions. The findings of this review further inform the selection of HR, HRV, and head movement parameters that have been used in previous studies to determine stress levels.

Chapter 4 provides a detailed description of the developed system. Chapter 5 describes the two experimental phases—an on-site phase and a week-long take-home phase—and explains the data collection process. Chapter 6 presents the experimental results, which are then discussed in Chapter 7. Finally, Chapter 8 outlines the thesis's contributions to the broader scientific field, acknowledges its limitations, and offers recommendations for future work.

1.5 Usage of generative AI

To improve grammar and readability large language models (LLMs) were used in this thesis to proofread and rewrite paragraphs to improve and unify the language used. The LLM output was always proofread by the thesis' author to ensure meaning and intention were not altered. The content and interpretations in this thesis are the work of the thesis's author. The thesis's abstract was automatically translated from English to German using ChatGPT-5.6 Sol.

2 Background

This chapter introduces the concepts necessary for understanding the remainder of this thesis. It begins by discussing physiological measurements in XR in Section 2.1 and continues by introducing the concept of stress in Section 2.2. It then examines how XR-based stress interventions are used to support caregivers and nurses in Section 2.3. Next, it introduces the Stroop effect, which is used as a mild psychological stressor in the application developed in this thesis, in Section 2.4. Finally, the chapter explains the relevant stress biomarkers, namely heart rate and heart rate variability (HRV), in Section 2.5, and head movement in Section 2.6.

2.1 Physiological measurements in XR applications

XR applications are increasingly used beyond entertainment-focused fields such as gaming (Ahmed et al., 2025). The use of physiological sensors in combination with XR systems is an active research field, with applications in psychology, medicine, sports, ergonomics, military, and tourism (Guillen-Sanz et al., 2024). Physiological sensors allow researchers to investigate phenomena that are difficult to assess using self-report methods alone (Grassini & Laumann, 2020; Halbig & Latoschik, 2021). Compared with subjective measurements such as questionnaires, interviews, or thinking aloud, physiological measurements can be acquired during the application and yield quantitative data that can be analyzed using machine learning (Halbig & Latoschik, 2021).

Across multiple studies, the literature reports a wide range of measures, including cardiac measures, electrodermal measures, electroencephalography (EEG), respiratory measures, electromyography (EMG), eye tracking, skin temperature, and movement-based signals (Ernst et al., 2024; Grassini & Laumann, 2020; Halbig & Latoschik, 2021; Wang et al., 2025). In XR research, cardiac and electrodermal measures are among the most commonly used because they can be acquired continuously and relatively unobtrusively while providing information about autonomic activation (Guillen-Sanz et al., 2024). Heart rate and heart rate variability are often used as indicators of stress, arousal, and cognitive workload, whereas electrodermal activity is commonly employed to assess changes in sympathetic nervous system activity (Ahmed et al., 2025). Other measures, such as EEG,

EMG, respiration, eye tracking, and skin temperature, can complement these signals by capturing neural activity, muscular responses, breathing behavior, visual attention, and peripheral physiological changes (Grassini & Laumann, 2020; Wang et al., 2025). In addition, movement-based signals such as head and body motion are increasingly relevant in XR because they can be obtained directly from the tracking system and may provide further insight into user state and behavior during immersive experiences (Holzwarth et al., 2021).

Overall, physiological sensing is valuable in XR because it enables the assessment of user state during the experience itself. This is particularly important in immersive applications, where stress, arousal, workload, attention, and discomfort may change rapidly over time and directly influence performance, interaction, and the overall user experience. Accordingly, the selection of measures depends not only on the construct of interest but also on practical considerations such as unobtrusiveness, measurement reliability, and compatibility with the XR setup (Guillen-Sanz et al., 2024; Halbig & Latoschik, 2021). In the context of this thesis, the focus is on acute psychological stress in XR and on measures that can be recorded continuously during immersion, namely heart rate, heart rate variability, and head movement.

2.2 Stress

Selye (1976, p. 137) defined stress as "the nonspecific response of the body to any demand made upon it". He further stated that stress does not need to be avoided at all times, as life itself causes temporary increases in stress and the complete absence of stress means death.

Looking at stress through a more modern lens, stress is how the body responds to external stimuli that pose an actual or perceived harm to our survival. Stress leads to either a fight, flight, or freeze reaction of the human body (Kirby et al., 2024).

The stress response is the body's way of restoring homeostasis and ensure survival. It includes a rapid activation of the sympathetic stress response and a slower hormonal response that raises cortisol levels. During the stress response, blood pressure, breathing rate and heart rate increase to make more energy available for fast physical and mental reactions. Processes that are not essential for immediate survival, such as saliva production and the digestive system are slowed down (Kirby et al., 2024; Russell & Lightman, 2019).

A stressor is anything that leads to a stress response. Stressors can be physical such as exposure to extreme weather, psychological like the frustration of losing one's job or social due to isolation or experiencing bullying (Kirby et al., 2024; Pinel et al., 2024).

This thesis focuses on the measurement of acute psychological stress in virtual reality

(VR), specifically participants' responses to a VR-based version of the Stroop task described in Section 2.4.

2.3 XR in stress interventions for nurses

Recent research positions virtual reality (VR) as a promising medium for stress support in nursing and related healthcare professions. In this context, VR is used less as a tool for clinical skills training and more as a means of creating immersive spaces for recovery, mindfulness, and emotional regulation. Existing interventions typically combine calming natural environments with guided relaxation, breathing, psychoeducation, or resilience-oriented exercises, and are delivered either as short sessions during work breaks or as repeated programs over time. A recent systematic review found that these interventions are generally feasible and well accepted in hospital settings, with evidence for reduced subjective stress and anxiety, improved mood, and positive effects on well-being and resilience-related skills (Haeussl et al., 2026). This broader picture is consistent with nurse-relevant primary studies.

Nijland et al. (2021) showed that ICU nurses used a brief immersive relaxation intervention during work shifts and reported markedly lower immediate stress afterward. Pallavicini et al. (2025) further indicated that similar intervention principles can also be transferred to self-guided home use by combining psychoeducational and relaxation-oriented content in a short remote VR program. Taken together, the literature shows that VR-based stress interventions for nurses are already being explored across relaxation, mindfulness, and resilience-oriented formats, and that immersive technology is increasingly seen as a practical tool for supporting recovery from occupational stress (Haeussl et al., 2026; Nijland et al., 2021; Pallavicini et al., 2025).

This context is directly relevant to the present thesis, which is part of the XR2esilience project on XR-based resilience training for nurses. While the broader project focuses on resilience interventions, this thesis develops the recording system needed to capture stress-related data during XR experiences and to support the later evaluation of these interventions.

2.4 Stroop effect

Stroop (1935) published a series of experiments on interference that introduced what is now known as the Stroop Color and Word Test (SCWT). In its original form, the test consisted of three tables that participants were asked to complete as quickly as possible. The

first two tables served as control conditions: in the first, participants read aloud color names printed in black ink, and in the second, they named the colors of printed patches. The third table represented the critical incongruent condition, in which participants were required to name the ink color of color words printed in incongruent colors. This condition gave rise to what is now known as the Stroop effect (Scarpina & Tagini, 2017).

In the SCWT, participants are typically instructed to respond as quickly and accurately as possible to a series of stimuli while ignoring task-irrelevant information. In the critical incongruent condition, they must name the ink color in which a color word is printed rather than reading the word itself. For example, if the word “blue” is presented in red ink, the correct response is “red.” Because reading is generally more automatic than color naming, the written word activates a competing response tendency, which slows responding and increases the likelihood of errors. Performance is therefore commonly assessed using response time, completion time, and error rate (Scarpina & Tagini, 2017). A stimulus list for the incongruent condition is shown in Figure 2.1.



Purple Brown Red Blue Green
Red Green Brown Purple Blue
Blue Brown Red Green Purple

Figure 2.1. Example schematic of the incongruent condition of the Stroop Color and Word Test. Participants are instructed to name the ink color of each word while inhibiting the automatic tendency to read the word itself.

The SCWT is an established acute psychological stressor in stress research (Cipresso et al., 2019; Renaud & Blondin, 1997). It has also been adapted into several XR implementations (Gradl et al., 2019; Morales Tellez et al., 2023; Ringgold et al., 2024), which appear to preserve the core interference processes of the Stroop paradigm while embedding the task in immersive and interactive environments and eliciting measurable physiological stress responses as well as subjective perceptions of stress in participants.

2.5 Heart rate and heart rate variability

Heart rate (HR) and heart rate variability (HRV) are important physiological indicators in stress research because they reflect autonomic regulation and change systematically in

response to stress and recovery processes (Blum et al., 2019; Shaffer & Ginsberg, 2017). One way to calculate HR and HRV is from an electrocardiogram (ECG) signal. An ECG records the heart's electrical activity over time by using electrodes placed on a person's skin to detect the small electrical potentials generated during each cardiac cycle. (*Electrocardiography (ECG)*, 2013)

The typical ECG curve consists of spikes and waves that are denoted as **P**, **Q**, **R**, **S**, and **T**. Phases **Q**, **R** and **S** are commonly grouped as the QRS complex (Köhler, 2022). Figure 2.2 illustrates a typical representation of these ECG components. The P wave reflects the electrical activation of the atria. The QRS complex represents the electrical activation of the ventricles and is usually the most prominent part of the ECG signal. The T wave reflects the recovery of the ventricles after contraction (*Electrocardiography (ECG)*, 2013).

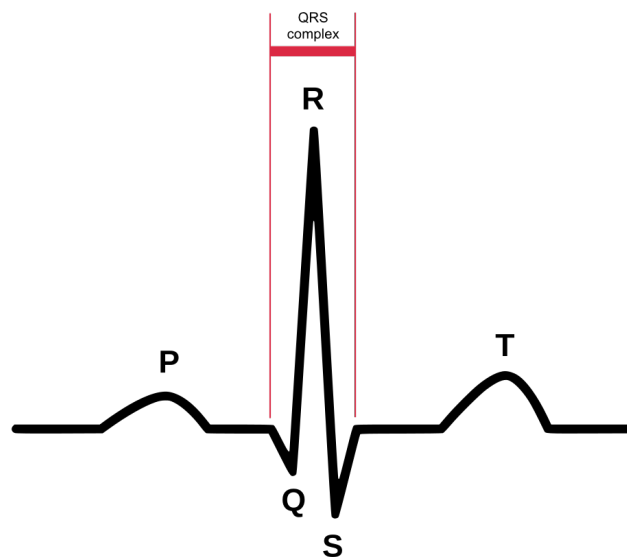


Figure 2.2. Electrocardiogram of a Heartbeat with P,Q,R,S and T labels. Note. Adapted from *ECG-P+QRSkomplex+T.svg*, by Agateller, 2008, Wikimedia Commons (<https://commons.wikimedia.org/wiki/File:ECG-P%2BQRSkomplex%2BT.svg>). CC BY-SA 3.0.

To derive HR and HRV from the ECG signal, the R peaks of successive QRS complexes can be detected. The time between two consecutive R peaks is called the R-R interval, which forms the basis of HRV analysis (Quigley et al., 2024; Task Force of the European Society of Cardiology the North American Society of Pacing Electrophysiology, 1996). Figure 2.3 illustrates the R-R interval. Heart rate can be calculated by converting the duration of these R-R intervals into beats per minute.

HRV describes the variation between successive R-R intervals and can be quantified using

different time-domain metrics. Two commonly used measures are SDNN, which reflects the overall variability in normal-to-normal intervals, and RMSSD, which captures short-term beat-to-beat variation and is closely related to parasympathetic cardiac activity (Shaffer & Ginsberg, 2017). In stress research, these metrics are relevant because changes in autonomic nervous system activity are often reflected in reduced HRV, whereas higher HRV is generally associated with greater physiological flexibility and self-regulation (Blum et al., 2019).

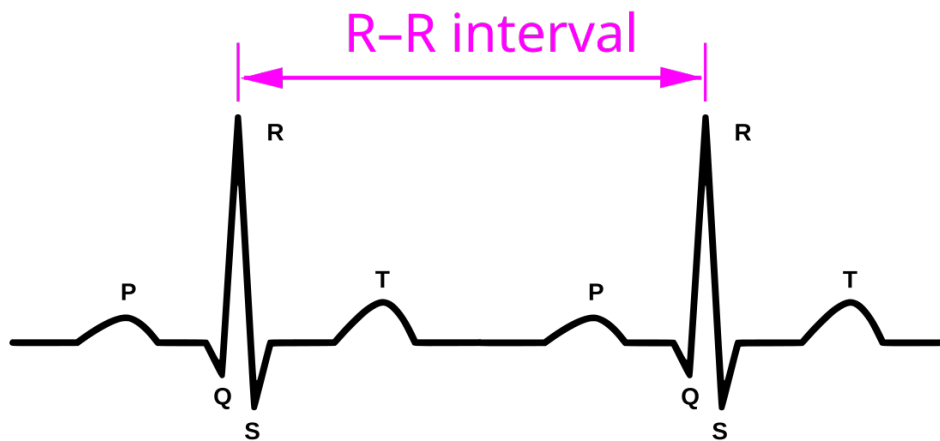


Figure 2.3. Electrocardiogram showing two periods and the RR-interval by Kychot, 2009, Wikimedia Commons (<https://commons.wikimedia.org/wiki/File:ECG-RRinterval.svg>). Copyrighted free use.

2.6 Head movement tracking in XR

XR systems must continuously estimate the user's head position and orientation and align the displayed virtual scene with the user's movements in order to present a convincing virtual scene (LaValle, 2019). The accuracy of this estimation directly affects the user's immersion and comfort while using the XR application. Systems providing low-latency head tracking are able to convey a feeling of "being there", often called presence, to users (Slater & Sanchez-Vives, 2016). Inaccurate tracking can reduce presence, impair inter-

action with virtual content, and, in some cases, contribute to discomfort or cybersickness (Slater, 2009).

Modern XR systems such as the Meta Quest 3 and PICO 4 Ultra support six degrees of freedom (6DoF) head-tracked experiences using onboard sensors and environment tracking (Meta Horizon Developers, 2026; PICO, 2024). This means that the system estimates orientation in terms of yaw, pitch, and roll, together with the head's 3D position in space. This is especially important for immersive applications, as it allows the system to mirror full head movements for rendering purposes. In the context of stress and resilience, this continuous stream of pose data can also be used for further analysis (Ferrarotti et al., 2024).

One important variant of 6DoF tracking is inside-out tracking, in which the XR headset operates without requiring additional external sensors (LaValle, 2019; Ranganathan, 2022). This approach is typically realized by combining inertial measurement units (IMUs) and cameras to solve the simultaneous localization and mapping (SLAM) problem (Leutenegger et al., 2014). The IMU consists primarily of a gyroscope, which measures angular velocity, and an accelerometer, which measures the acceleration acting on the headset. Using IMU data allows the system to predict changes in motion with an associated error term (Wetzstein, 2023). This error accumulates over time, eventually causing a noticeable discrepancy between the predicted motion and the actual movement. To counteract this drift, the headset uses its integrated cameras to estimate landmarks in the environment as 3D points that correspond to features in the real world (Ranganathan, 2022). An image showing such landmark points can be seen in Figure 2.4.

By using multiple cameras and images captured at different points in time, while the approximate motion is known using the IMU, the system is able to correct accumulated IMU errors at a known interval (Ranganathan, 2022). The Meta Quest 3 HMD captures tracking images either at 50 Hz or 60 Hz, depending on the environment.

In this thesis, head movement data are additionally recorded to investigate whether and how movement parameters relate to stress. Because these data are obtained directly from the XR tracking system, they provide a non-invasive movement-based indicator of user state during VR exposure.



Figure 2.4. Example of Quest detecting landmarks in the environment, by A. Ranganathan, 2022, Source (<https://www.aiacceleratorinstitute.com/the-oculus-insight-positional-tracking-system-2/#slam-computer-vision-method>)

3 Literature Review

3.1 Goal of the literature review

This literature review aimed to identify and synthesize studies that measured stress in immersive VR/XR settings using heart rate, heart rate variability, and head movement data. A literature review was conducted, informed by the PRISMA guidelines (Page et al., 2021).

3.2 Search Strategy

To identify relevant studies that use head movements to measure stress in VR the repositories were queried with the search string “Stress AND Virtual Reality AND head movements.”

To find relevant studies that use heart rate variability (HRV) and heart rate for stress measurements, the repositories were searched using the search string “Stress[Title] AND heart rate AND Virtual Reality.”

Repository	Search String	Results
ACM Digital Library	[Title: stress] AND [All: virtual reality] AND [All: head movement]	103
ACM Digital Library	[Title: stress] AND [All: heart rate] AND [All: virtual reality]	108
PubMed	((Stress[Title]) AND (Heart Rate)) AND (Virtual Reality)	112
PubMed	((Stress) AND (Virtual Reality)) AND (head movement)	16
IEEE Xplore	("Document Title":Stress) AND ("All Metadata":Heart Rate) AND ("All Metadata":Virtual Reality)	69
IEEE Xplore	("All Metadata":Stress) AND ("All Metadata":head movement) AND ("All Metadata":Virtual reality)	44
CINAHL	TI (Stress) AND heart rate AND virtual reality	25
CINAHL	Stress AND head movement AND virtual reality	1
Springer Nature Link	Stress[Title] AND Virtual Reality AND heart rate	238
Springer Nature Link	Stress[Title] AND Virtual Reality AND head movement	87
ScienceDirect	Virtual Reality AND Heart Rate; Title: Stress	370
ScienceDirect	Virtual Reality AND head movement; Title: Stress	157
Google Scholar Labs	I am looking for studies that measure stress in VR applications using heart rate and heart rate variability	20
Google Scholar Labs	Find studies that measure stress in VR applications using head movements	10

Table 3.1. Table describing searched repositories, used search queries and number of results during systematic literature review.

1360 records were identified before deduplication, and 882 remained after duplicates were removed.

3.3 Screening

During screening the titles and abstracts were assessed to decide which records to exclude. The exclusion criteria were:

- Result not about XR
- Result does not include biomarkers

- Result not about stress
- Result is meta-analysis or review
- Result not about humans
- Result does not include Heart Rate or Head Movements
- Was not able to determine how to access result

After screening titles and abstracts 707 records were excluded and 175 records were assessed for eligibility by reading the full text. The exclusion criteria were:

- Result does not focus on XR enough
- Result does not include HRV or head movements
- Result does not focus on biomarkers enough
- Result is meta analysis or review
- Result does not conduct experiment
- Result is not in-depth enough
- Sample size of experiment too small
- The full text was not accessible

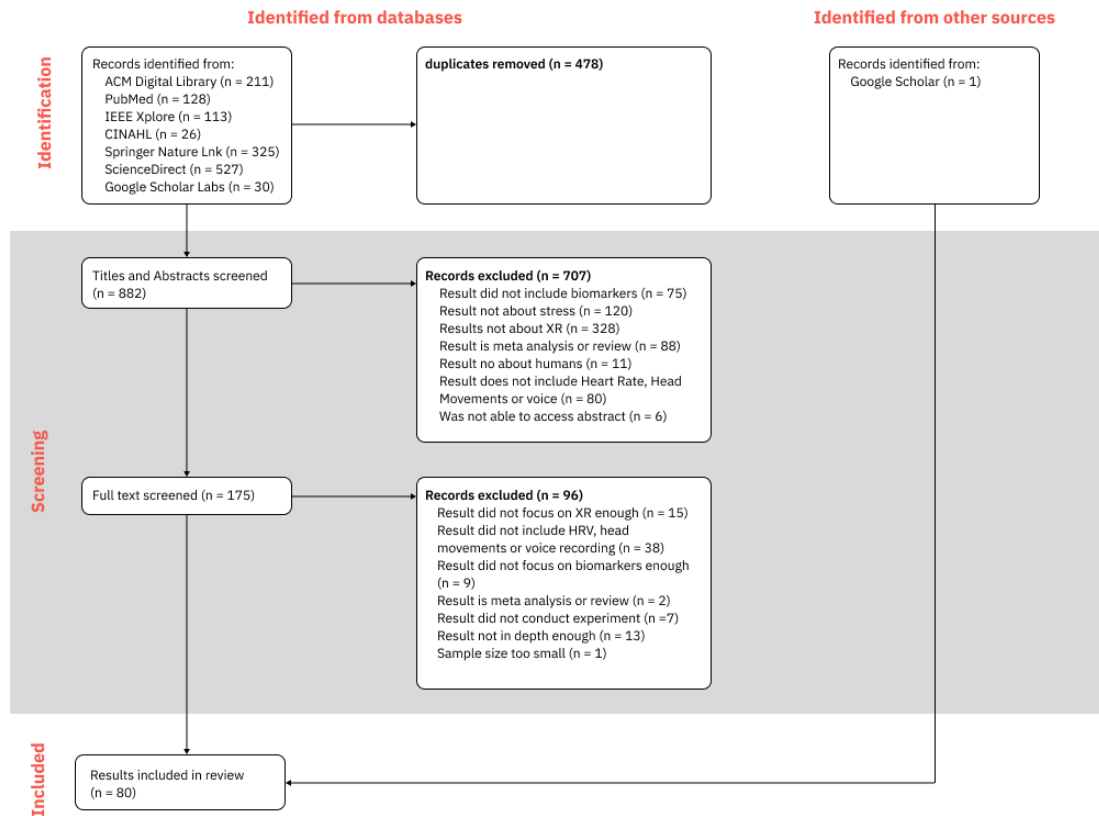


Figure 3.1. Flow diagram according to Page et al. (2021) depicting review process

3.4 Review

After the screening process, 79 records were selected for inclusion in the literature review. ResearchRabbit (<https://www.researchrabbit.ai/>) was used to support the identification and selection of relevant and impactful publications. In this section, a subset of these works is discussed in greater detail.

3.4.1 VR and biofeedback for stress management

Previous works explored the usage of biofeedback integrated in VR interventions. Weibel et al. (2023) conducted a study on 115 participants randomly assigning each participant to one of four conditions. The conditions were HRV biofeedback via VR, HRV biofeedback via desktop screen, standardized paced breathing via VR, and standardized paced breathing via desktop screen.

Weibel et al. (2023) implemented a 2x2 between-subject study comparing HRV biofeed-

back versus standardized paced breathing delivered either via an Oculus Quest HMD or a 2D desktop display. The VR application was built in Unity and presented a nature scene with an animated visual breathing pacer. Cardiac activity was acquired using a Polar H10 chest strap; data were streamed to a custom Windows app (Polar SDK) and forwarded to a centralized database for RF estimation and feedback computation. Real-time processing used Python (hrv v0.2.8; scipy v1.5.3) with artifact/ectopic filtering and FFT-based PSD from RRs interpolated at 4 Hz. For HRV-BF, resonance frequency was estimated during an initial 6-min block and used to pace breathing in two subsequent 10-min training blocks; biofeedback was based on a coherence-ratio-derived score updated every 10 s (90 s window) that drove discrete VR changes (e.g., flower growth, sun position, sound intensity). In the sPB condition, training blocks used a fixed 0.1 Hz pacer and the environment changed independent of physiology.

The setup did not show any significant differences in psychological or cardiac measures between conditions in which VR was used and screen setups, nor did it show significant differences between biofeedback and standardized paced breathing conditions.

Daniel-Watanabe et al. (2025) describe two studies in which they investigate the impact of prior VR biofeedback training on participants' performance in an immersive VR game in which the participants had to control their heart rate to avoid losing the game. The game simulates the player being in a dungeon where they have to avoid being detected by a blind monster that can hear their heart rate. Heart rate related biomarkers were recorded with a BioRadio measurement device. For measuring the heart rate variability (HRV), a PPG sensor worn on the participant's non-dominant hand was used. The authors used the standard deviation of N-N intervals (SDNN) as an HRV metric. The second study showed that a group that had previously received VR based biofeedback training had higher SDNN values than the control group, indicating that VR based stress training can be an effective method for helping individuals cope with stressful situations.

3.4.2 Biophilic VR experiences related to stress

Yin et al. (2020) conducted a between-subject study with 100 participants. The participants were first placed in a stressful environment people were first put in a stressor environment where they had to perform two stress inducing tasks before being randomly assigned to one of four virtual recovery conditions with differing numbers of biophilic elements for six minutes. An HTC Vive VR headset was used as head mounted display (HMD) and physiological data were collected using a Movisens EcgMove3 (Movisens GmbH) using single-channel ECG to calculate heart rate (HR) and heart rate variability (HRV). In this study, HRV was calculated using the root mean square of successive differences between normal heartbeats (RMSSD) and low to high frequency ratio (LF/HF). The study's results

provided evidence that VR rooms with biophilic indoor designs had a greater effect on reducing physiological stress during recovery than the other conditions, while a simulated view of the outdoors had the greatest effect on reducing anxiety.

You et al. (2023) describe a similar study with 30 participants. The participants were exposed to five different virtual classroom settings (four biophilic, one non-biophilic control). The order of the five settings was randomized for each participant. The study used the HTC VIVE CE headset and SteamVR as a tracking system. Blood pressure (BP), heart rate (HR), heart rate variability (HRV), and skin conductance level (SCL) were measured. For measuring HR and HRV they used a Shimmer3 GSR+ device positioned at the subject's left earlobe to collect photoplethysmography (PPG) data. To determine HRV, the root-mean-square of successive differences between normal heartbeats (RMSSD) was used. This study did not find a significant effect of biophilic interventions on physiological stress indicators. The authors claim that subjects reported significantly lower perceived stress and a stronger connection with nature in biophilic virtual environments.

Serra et al. (2025) conducted a study with 41 participants. 21 participants were assigned to a real office environment and 20 participants were put in a virtual copy of the real environment. During the sessions, the participants' physiological responses were measured with an Empatica E4 wristband. The wristband recorded blood volume pulse (BVP) at a rate of 64 Hz, from which the device automatically calculated inter-beat intervals (IBIs). The IBIs were then used to derive HRV which was then normalized to %HF. Each participant completed a five-minute baseline in a neutral environment followed by a backward digit span task in the same neutral environment. After the baseline, the participants were exposed to a living wall or a non-living wall. This exposure was done in VR for the VR group and in a real environment for the non-VR group. Seven days later they performed a second session with the participant being exposed to the other condition. This meant that participants who had previously been exposed to a non-living wall were exposed to a living wall in the second session, and vice versa. The study reports that the backward digit span induced physiological stress as measured by %HF. When the participants were exposed to the living wall condition they had significantly higher %HF compared to the non-living wall. These effects were observed in both real and VR conditions with no significant differences between VR and non VR groups.

Taken together, these studies suggest that biophilic XR environments can support stress regulation and that HRV is a useful biomarker for assessing these effects. Yin et al. (2020) showed improved physiological stress recovery in biophilic VR settings, while You et al. (2023) found stronger effects on perceived stress than on physiological markers. Serra et al. (2025) further showed that biophilic elements can reduce physiological stress during demanding tasks, with similar effects in real and virtual environments. Overall, the studies indicate that HRV-based measurement is feasible for XR resilience interventions, but

outcomes depend on study design, measurement quality, and intervention context.

3.4.3 Stress prediction from head movement in XR

Kusano et al. (2020) propose a machine learning approach for predicting subject's stress levels based on their head motions. They extracted heartbeat-like features from head movement using a procedure previously described by Hernandez et al. (2014). In their experiment Kusano et al. (2020) tested 35 subjects on the Ball Control Task and Berg Card Sorting Task. Furthermore, they tested 38 subjects on movie watching tasks. The data collection was done with an Oculus Rift CV1 VR headset and eMotion Faros 360 ECG sensor. Head motions were measured as angular acceleration at a frequency of 600 Hz. The study showed that the model achieved similar accuracy when using the derived features from the head motions as when using ECG measures directly for classifying whether the data originated from a stressful recording or a non-stressful one.

Ferrarotti et al. (2024) explored using head movement for stress prediction by utilizing the Microsoft HoloLens 2 AR headset. 100 participants were exposed to an AR version of the Stroop Color Word Test (SCWT). 20 of the 100 participants additionally performed mental Arithmetic tasks. They recorded the headset position together with a timestamp during the experiment in order to analyze head movements later. The SCWT was conducted in three phases. The third phase consisted of the incongruent condition and was labeled as stress inducing during data analysis. The study did not measure physiological stress. Their results show that they were able to correctly classify the stressful condition of the test at 98.33% accuracy using a Total displacement metric derived from the recorded positions using an SVM classifier architecture.

3.4.4 Stroop Test

Gradl et al. (2019) developed a modified version of the Stroop test called Stroop Room that is adapted to provide a 360° challenge in order to make use of the immersive properties of virtual reality (VR) technology. They evaluated the Stroop Room on 32 subjects. An HTC Vive headset was used by employing the SteamVR library for showing the VR content. To collect physiological biomarkers, the study used a Nexus Kit 10 MKI to collect electrocardiogram (ECG) and electrodermal activity. The study analyzed heart rate (HR), root-mean-square of successive differences between normal heartbeats (RMSSD), percentage of successive NN intervals that differ by more than 50 milliseconds (pNN50), and low frequency to high frequency ratio (LF/HF) as heart rate variability (HRV) measures. The study reports increased heart rate during both congruent and incongruent versions of the Stroop test. Furthermore, RMSSD and pNN50 values were lower in the incongru-

ent trial compared to the congruent trial and substantially increased LF/HF values. These results support the claim that the Stroop Room induces cognitive stress.

Morales Tellez et al. (2023) compared two VR versions of the Color-Word Stroop test (CWST) to a conventional implementation of the test without any VR environment. They collected the Stroop-Color score of the study participants and used a Polar H10 band to collect heart rate data. Additionally, they collected perceived stress by employing the DASS-21 questionnaire before and after the sessions. Their results showed that there were no significant differences in Stroop scores between the VR and non-immersive versions of the test, but the average heart rate of study participants is higher under the VR conditions. The perceived stress scores were similar between all conditions.

Ringgold et al. (2024) performed a validation of the Stroop Room introduced by Gradl et al. (2019). Their sample consisted of 89 participants that were evenly distributed across the three conditions of the original Stroop Room. An HTC Vive Pro Eye headset with one controller was used via SteamVR as a VR system. ECG was recorded with a Nexus Kit 10 MKI, copying the setup from the previous study. In their results, participants showed significantly higher HR during the incongruent phase of the Stroop Room. In this validation study the authors did not record and process HRV related variables unlike the original setup.

These three studies suggest that VR-based implementations of the Stroop test are able to preserve core cognitive properties of the conventional Stroop task. All three studies show measurable physiological results and across all three studies performance on the Stroop task remained comparable to non VR based implementations. All three studies report higher HR during the incongruent condition, and Gradl et al. (2019)) reports effects on the participants' RMSSD and pNN50 measures.

3.4.5 Conclusion

Overall, the reviewed studies indicate that VR is a viable technology for stress research, as it enables controlled stress induction, immersive intervention design, and multimodal physiological assessment. The findings suggest that VR can reproduce established stress paradigms and support meaningful stress-related effects. Among the physiological measures used, HRV appears particularly informative, and ECG-based acquisition seems to offer the most robust basis for capturing autonomic stress responses. Furthermore, early evidence suggests that head movement may serve as an additional unobtrusive source of stress-related information in XR, although further validation is needed before it can be considered a reliable standalone indicator. Taken together, the literature supports VR as a strong methodological fit for stress studies and highlights ECG-based HRV as a promising core measure for future work.

4 Application and Apparatus

This chapter describes the hardware setup used during the user tests described in Chapter 5, as well as the developed library and Unity application.

4.1 Hardware

This section describes the hardware used for this thesis. It provides a brief overview of what each component is and why it was chosen.

4.1.1 Meta Quest 3

This thesis uses a Meta Quest 3 head-mounted display (HMD) as the primary hardware platform. The Meta Quest 3 is a standalone virtual reality headset that integrates onboard processing, inside-out tracking, and controller-based interaction in a single device, allowing VR applications to be executed without the need for an external computer or fixed tracking infrastructure. It was selected for this study primarily because its standalone operation allows for usage in take home settings and reduces external dependencies during data collection. In addition, the Quest platform is one of the most widely adopted contemporary consumer VR ecosystems, with broad developer support and widespread availability, making the Meta Quest 3 a practical and relevant choice for applied VR research (Meta, n.d.).

4.1.2 Polar H10

This thesis uses the Polar H10 HRM (Polar Electro Oy, Kempele, Finland) to record cardiac activity during the experimental sessions. The Polar H10 is a commercially available chest-strap sensor that captures beat-to-beat R–R intervals via electrode-based measurement at the thorax, making it suitable for heart rate and heart rate variability (HRV) analysis in both laboratory and ambulatory settings. It was selected for this study because it combines practical usability, low setup complexity, and broad commercial availability with measure-

ment quality that has been shown to closely approximate electrocardiogram (ECG)-based reference systems.

This makes the Polar H10 a particularly suitable choice for applied stress and resilience research, where robust physiological recordings are needed without the complexity of full clinical ECG instrumentation. Previous validation work has shown strong agreement between the Polar H10 and ECG for linear HRV parameters during resting and exercise conditions (Schaffarczyk et al., 2022). In addition to validation studies, the Polar H10 has also been used in related applied settings such as virtual-reality-supported stress management, where it was employed to derive HRV biofeedback and to assess stress-related physiological changes (Kerr et al., 2023; Weibel et al., 2023). Taken together, these findings support the Polar H10 as a robust and widely used consumer-grade sensor for psychophysiological studies on stress regulation and resilience.

4.2 Software Architecture

A custom Unity package and application were developed to enable standalone VR data collection without peripheral computing requirements. Although the Meta Quest 3 served as the primary development hardware, the implementation relies on OpenXR and the XR Interaction Toolkit to support broad hardware interoperability. This section outlines the technical architecture, emphasizing a design that is both extensible and easily integrated into subsequent studies.

This section first introduces the custom-developed Android library in Subsection 4.2.1. It then describes the Unity package for on-device biomarker collection in Subsection 4.2.2. Finally, it outlines the specific implementation used to conduct the experiments of this thesis in Subsection 4.2.3.

4.2.1 Custom Android library

For heart rate related data collection with a Polar H10 heart rate monitor, a custom Android library has been developed using the Kotlin programming language.

The library scans for Bluetooth devices which implement the Heart Rate Service (Bluetooth SIG, 2011). When connecting to a device, the library determines if the connected device is a Polar H10 or an unspecified heart rate monitor (HRM). If the connected device was identified as a Polar H10 the library allows for ECG data streaming and reports RR intervals to the host application. For ECG data streaming the library uses the proprietary Polar Measurement Data (PMD) service. If the device type cannot be determined the library only provides heart rate data.

The Unity host application uses the library as an `AndroidJavaClass` and provides a callback object that implements the `UnityCallback` interface for communication with the application.

4.2.2 Data collection package

The data collection package provides an extendable structure for managing the collection of biomarkers and answers to Likert scale questions to VR applications in Unity. The package is organized around the `RecorderManager` class, implementing the `IRecordingStorageManager` interface. The `RecorderManager` is provided to the different recorders via dependency injection. An overview of the package's architecture can be seen in Figure 4.1.

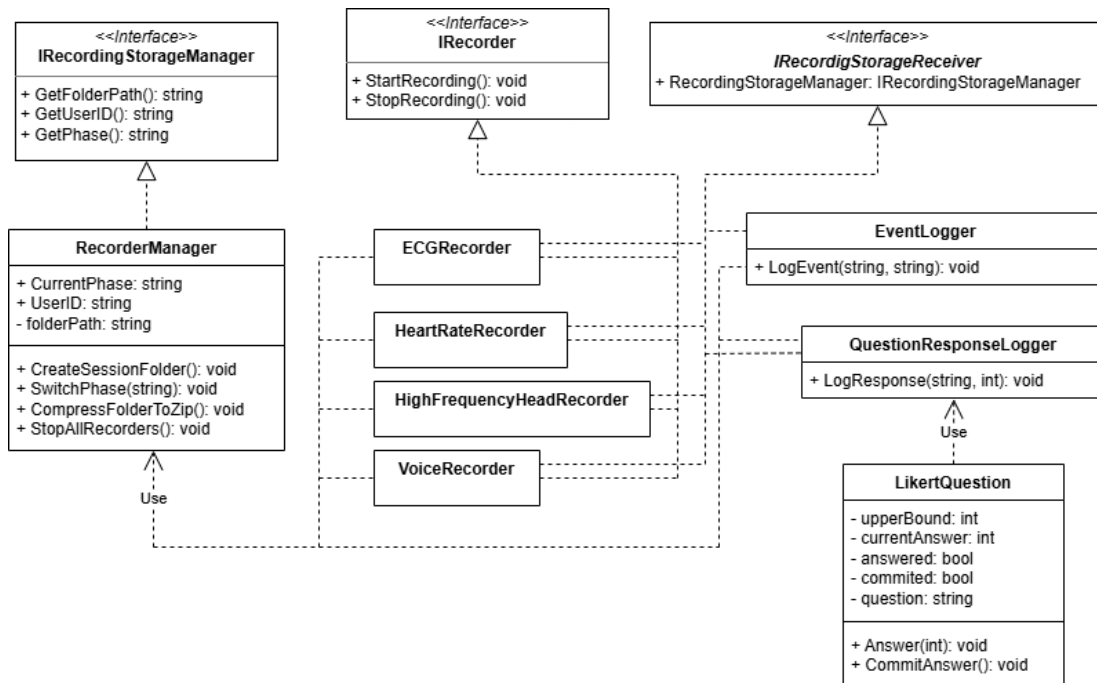


Figure 4.1. UML diagram describing an overview of the data collection package software architecture.

The recorders implementing the `IRecorder` and `IRecordingStorageReceiver` interfaces are responsible for handling how to collect and store the data for which they are responsible. `ECGRecorder` and `HeartRateRecorder` use the library described in Subsection 4.2.1, whereas the `HighFrequencyHeadRecorder` implements a `HandleIncomigData(InputEventPtr eventPtr)` method that reads the values from the head mounted display (HMD). This ensures that new head movement data is recorded whenever the headset provides it rather than waiting on the next rendered frame.

Additionally, there are the `EventLogger` and `QuestionResponseLogger` classes that im-

plement the `IRecordingStorageManager` interface. `QuestionResponseLogger` class allows for storing answers to Likert scale questions. In the experiment setup used for this thesis, this is used to store the participants' perceived stress level on a scale from one to five before and after the intervention. The `EventLogger` class is used to record phase switches and the participants' performance in the Stroop test section of the experiment. The package is designed to allow for reusing and repurposing the loggers in future projects.

Recorded data is stored on the HMD in a compressed zip file with a filename containing the specified User ID and the timestamp at which the session folder was created. `ECGRecorder`, `HeartRateRecorder`, and `HighFrequencyHeadRecorder` store their respective data in a CSV file with each row always containing the timestamp and current phase of the recording. `VoiceRecorder` stores voice recordings as WAV files. In this thesis it is used to collect verbal feedback from the participants on the application. `QuestionResponseLogger` and `EventLogger` store data in JSON files with each entry always containing a timestamp.

The software architecture is designed so that the package is reusable in future studies which require recording of biomarkers in VR applications and can be extended with classes implementing the `IRecorder` and `IRecordingStorageReceiver` interfaces to record data required for the specific study.

4.2.3 Experiment implementation

In this thesis, the library and package described in Subsections 4.2.1 and 4.2.2 were used to develop the experiment application to investigate the collection of biomarkers on device in a stress related intervention. The intervention design was informed by the requirements outlined by Wanner and Leung (2025). In this subsection the phases of the experiment application are described in detail.

Introduction and Connection

The participant starts the experiment in a meadow environment and is presented a UI giving context on the experiment. The participants needs to navigate through subsequent UI screens by using a Ray interaction and using the controller trigger to activate a button. This is to familiarize the participant with UI interactions in the VR environment. A screenshot of the first screen from the perspective of the user can be seen in Figure 4.2.

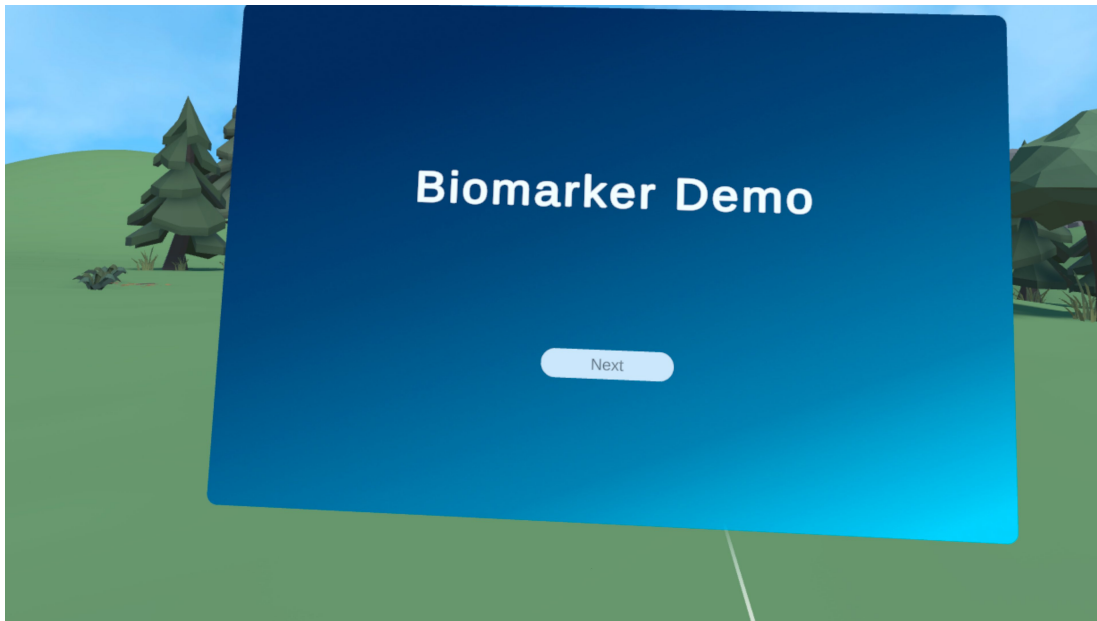


Figure 4.2. Screenshot of first menu screen of the application

After the introductory information the user will navigate to a screen on which they are presented with a reminder of how the HRM should be worn and how to connect it to the application. The user is presented with a list of buttons containing the device name and address and can start the connection by interacting with the button. This menu is presented in Figure 4.3.

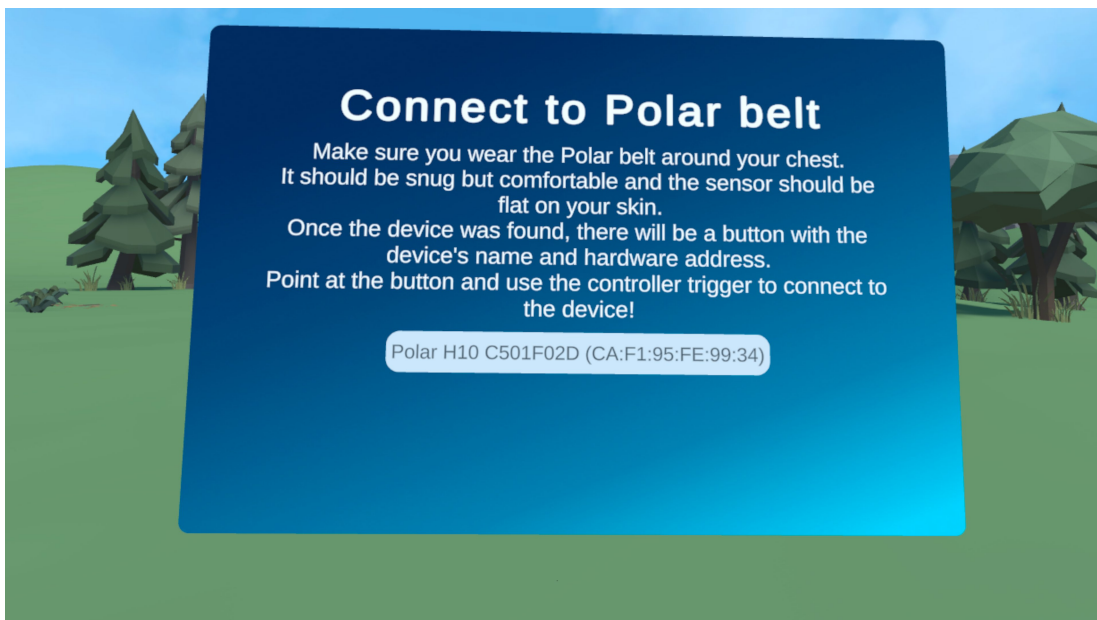


Figure 4.3. Screenshot depicting the "Connect to device" menu

When the user connects to the HRM the application switches to a screen reporting the currently measured heart rate, the HRM device's battery status, and a button to start the recording session. The battery status and current heart rate are shown to check that data is available and that the device has enough battery charge to record the full session. Figure 4.4 shows that screen.

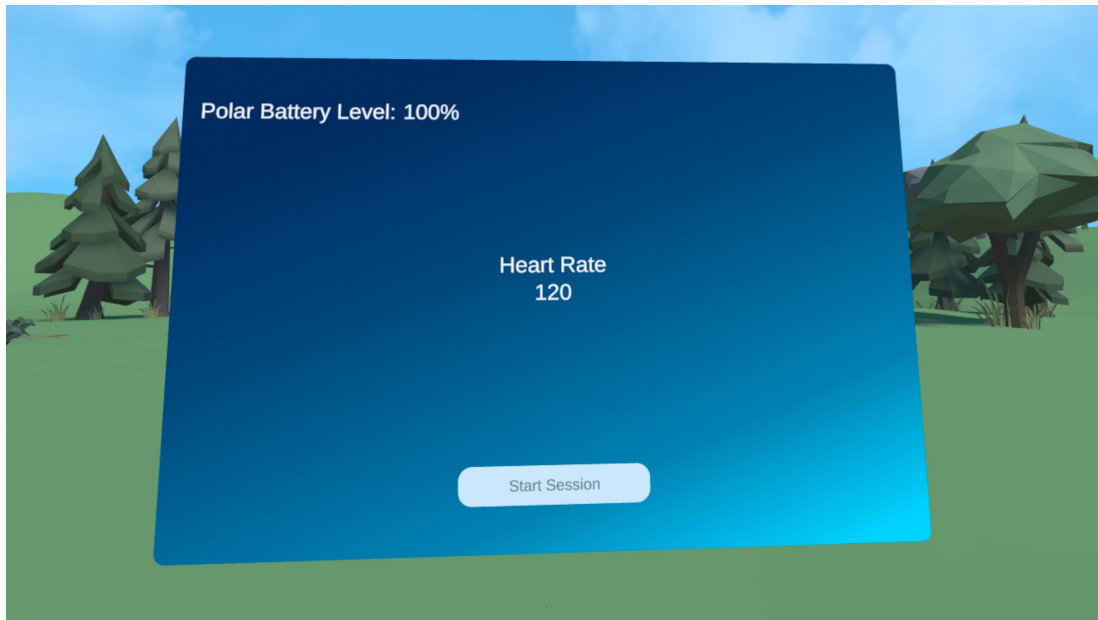


Figure 4.4. Screenshot depicting the menu showing heart rate and device battery

After the connection between headset and HRM device has been established, the participant's baseline is recorded for 90 seconds. During the baseline recording the participant is instructed to relax and the process is explained with a UI panel as shown in Figure 4.5. Following the baseline recording the user is presented with the question: "How stressed do you feel?" as shown in Figure 4.6. Their answer is recorded on a five point Likert scale from not stressed to extremely stressed. Answering the question concludes the first phase of the application and switches to the next scene.

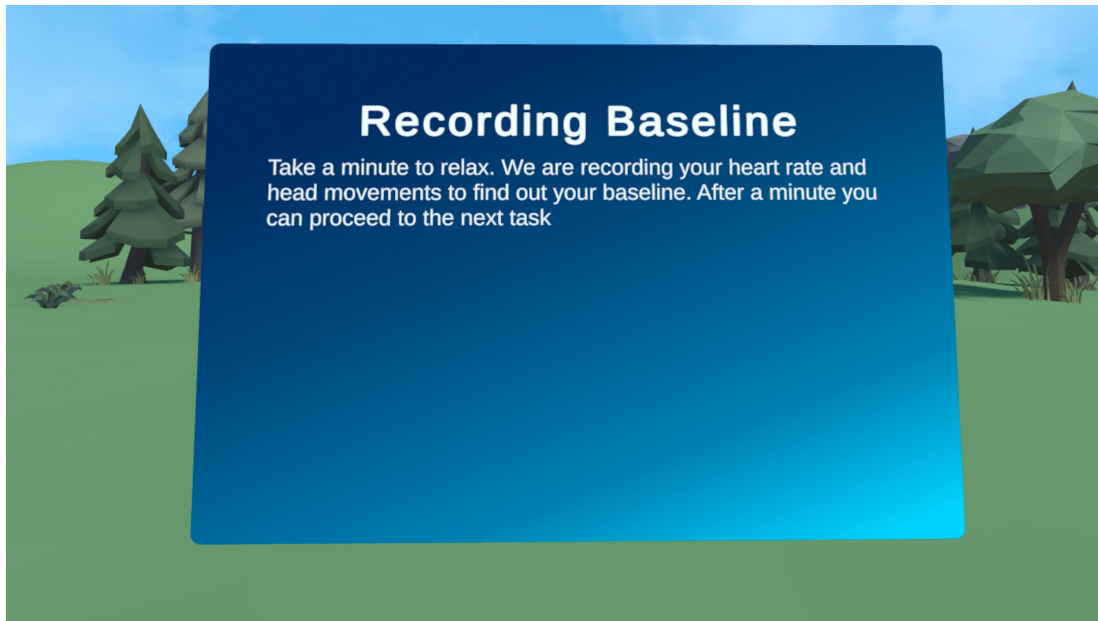


Figure 4.5. Screenshot depicting the instruction showed during baseline recording.

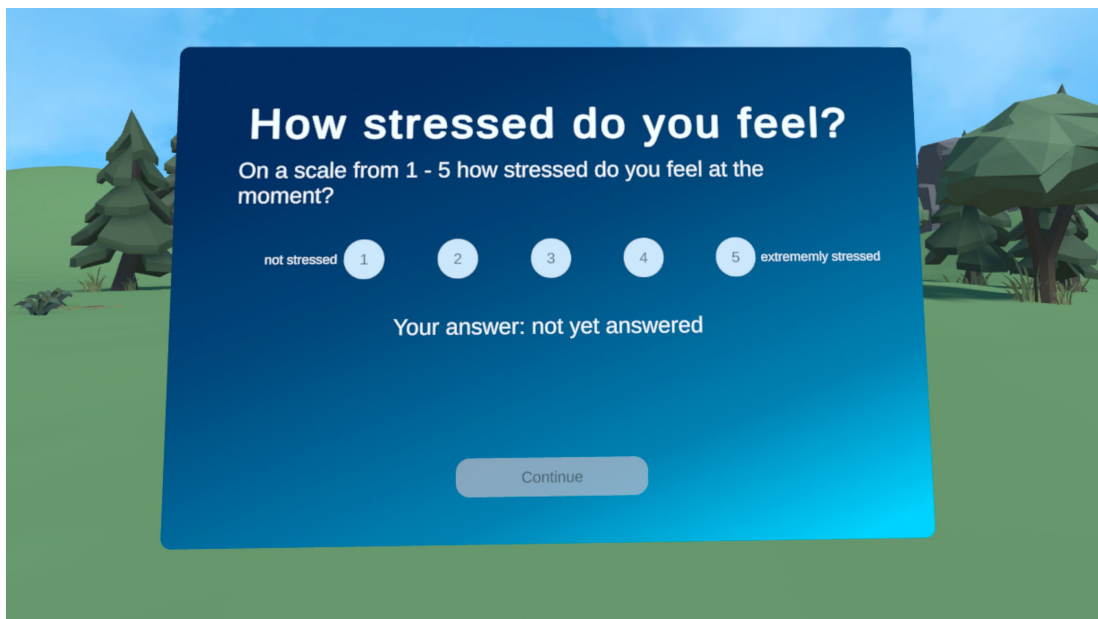


Figure 4.6. Screenshot depicting the question of "How stressed do you feel?" and the interface to answer the question

Stroop Room

The second phase takes place in a Stroop room scene built according to the description by Gradl et al. (2019). The room is hexagonal, with walls that are 10 meters wide and 3.5 meters high. The walls are arranged at a radius of 8.5 meters around the scene origin. At

the center of each wall is a color panel measuring 8 meters in width and 3 meters in height. This smaller color panel is a modification of the original layout, in which the color panels had the same dimensions as the walls. The panels were reduced in size and separated by a gap to prevent users from accidentally selecting a different wall than intended due to controller inaccuracies. A top-down view of the room layout is depicted in Figure 4.7.

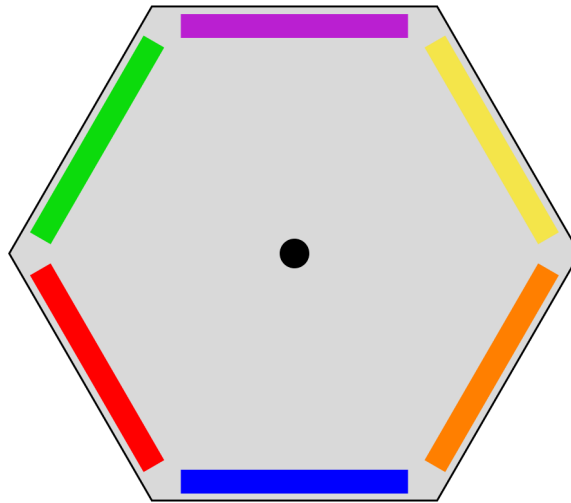


Figure 4.7. The wall layout of the Stroop Room scene depicted from above. The scene origin is illustrated with a black dot.

Before starting the Stroop test, the test rules are explained in a menu panel, as illustrated in Figure 4.8. During the test, the participant is presented with three panels positioned around the origin, each containing two lines. The first line can either be “Word” or “Color”, written in a white font on a dark gray background. The second line can be “red,” “green,” “purple,” “yellow,” “orange,” or “blue,” written in a randomly selected color from six colors. The last element on the panel is a green bar indicating how much time is left to complete the challenge. An example of this can be seen in Figure 4.9.

Each challenge is limited to 5 seconds. For every challenge, the participant must first read the word displayed in the first line and determine the required response rule. If the first line shows “Color,” the participant must ignore the semantic meaning of the second word and instead select the wall corresponding to the color in which the second line is written. If the first line shows “Word,” the participant must ignore the font color and select the wall corresponding to the color named by the second word. To make a selection, the participant points at the corresponding wall using the controller and confirms the choice by pulling the trigger. After each response, auditory feedback is provided to indicate whether

the selection was correct or incorrect, and the next challenge is presented immediately. This process continues until the overall test duration of 180 seconds has elapsed.



Figure 4.8. Screenshot depicting the explanation of the Stroop test rules.

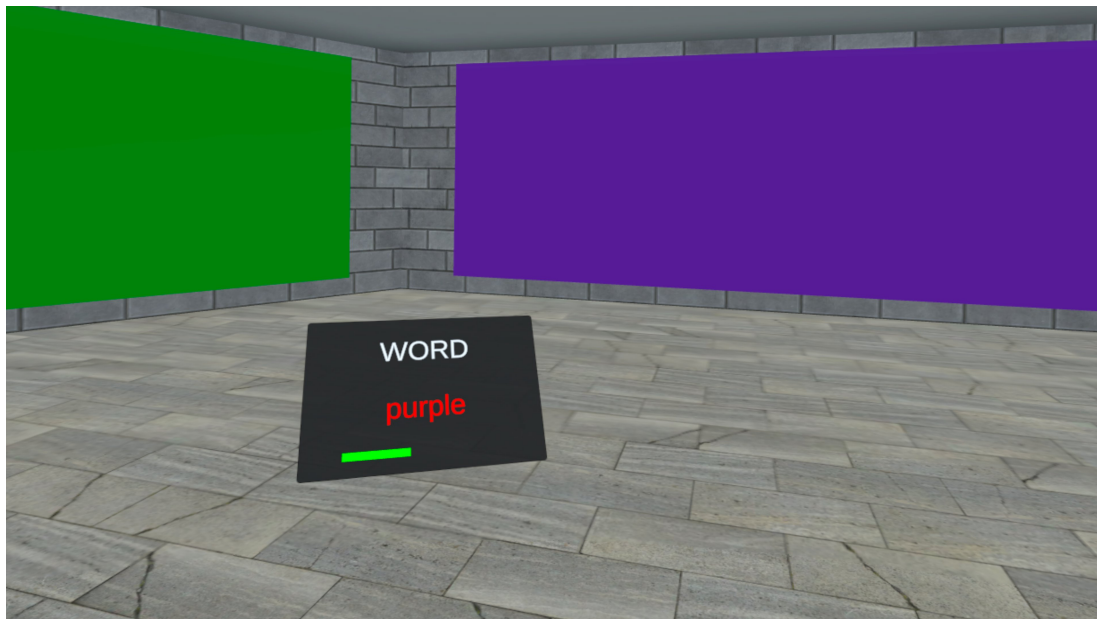


Figure 4.9. Screenshot from the user's perspective during the Stroop room test with the panel depicting the current challenge visible.

After the test is concluded, the user is presented with a results screen where they see how many challenges were presented in total and how many of those were answered correctly. Pressing the "Next" button on that screen transitions to a second iteration of the five point

Likert scale question: “How stressed do you feel?”. This is done to collect data on the user’s perceived stress level after the Stroop test. On the next screen the user is asked to record verbal feedback on the application as shown in Figure 4.10.

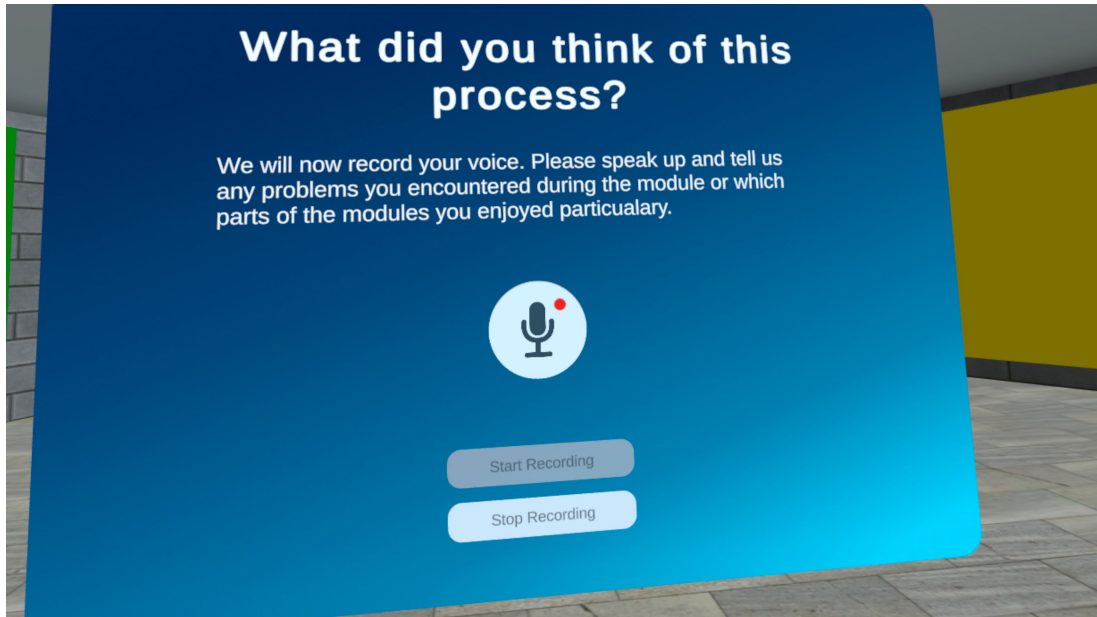


Figure 4.10. Screenshot depicting the screen asking the user to provide verbal feedback.

After providing verbal feedback, the application switches back to the meadow scene to record a 90-second cooldown phase to see how the participant’s biomarkers change after the stress intervention.

5 Method

This chapter outlines the methodology used to evaluate the library and the application presented in Chapter 4. It describes the study design, participant recruitment, test environment and setup, as well as the procedures used for data collection and analysis. The evaluation comprised two complementary study components, which are described separately in this chapter: a supervised on-site evaluation conducted at the University of Applied Sciences St. Pölten, and an exploratory one-week home-based evaluation completed independently by participants.

5.1 On-site evaluation

This section describes the performed on-site evaluation of the presented application. It first describes the study design and presents the design process including pretests. Then the procedure to evaluate the collected data is described. Finally, the recruitment strategy and recruited population for this evaluation phase is described.

5.1.1 Study design

To evaluate the developed data collection library and application introduced in Chapter 4 an exploratory, non-randomized feasibility and usability evaluation was performed. The first part of the evaluation consisted of a supervised on-site test session. The tests were conducted in the XR-DEV-Lab, a room at the University of Applied Sciences St. Pölten that is specifically designed for conducting VR tests. The study design used the same conditions for all participants and therefore did not include randomization or control groups. The study did not aim to evaluate clinical effectiveness or provide medical interpretations of the recorded biomarkers, only to evaluate the application's ability to record biomarkers relevant in stress research.

The study protocol design was informed by the recommendations outlined by Wanner and Leung (2025) and additionally reviewed by Sophia Wanner who provided crucial feedback on the content presented during the setup and baseline-recording phases, as well as on the appropriate duration of the individual phases of the application. Additionally, the application

was informally tested by two students of the “Interactive Technologies” master program who provided feedback on UI elements and the connection process between application and the Polar H10 heart rate monitor. None of the people involved during the study protocol design were included in the recordings for the final dataset.

During the on-site session participants were first informed about the session procedure both verbally and in writing via the informed consent form attached in appendix A. As part of the informed consent form, they were also asked to provide their age and gender as well as their name so that data subject rights can be upheld according to GDPR. After the initial explanation, participants were asked to put on the Polar H10 chest strap and depending on each participant’s familiarity with VR systems an explanation on how to use the Meta Quest 3 VR headset was given. The participants then started the application described in chapter 4 and proceeded to follow the instructions given by the application. During the session, participants were observed so that the researcher could provide assistance or intervene if necessary because of safety concerns. After using the application the participants were offered a short break and a glass of water, as well as salted crackers. After the short break a System Usability Scale (SUS) questionnaire as introduced by Brooke (1996) was administered to collect data on the application’s usability.

5.1.2 Evaluation procedure

In this subsection the steps to evaluate the data collected by the application described in Chapter 4 are described. This subsection first explains the evaluation of the System Usability Scale (SUS) (Brooke, 1996), then explains the evaluation of HRV features and lastly explains the evaluation of head-movement data.

SUS evaluation

The collected SUS questionnaires were evaluated according to the standard protocol described by Brooke (1996). Verbal audio recordings were transcribed and manually analyzed by the author to synthesize participant feedback on the application’s usability and to identify recurring usability issues. As described in chapter 4 participants were prompted to answer “How stressed do you feel?”, before and after the Stroop Room test in order to gather measurements about participants’ perceived stress levels. The answers to the question were analyzed to find if the participants perceived a change in stress level by calculating the differences of the answers. Additionally, the answers were used to investigate the relationship between recorded biomarkers and participants’ stress perception.

HRV evaluation

Using an in-house developed pipeline (Iselcor, 2026), heart-rate-variability (HRV) features were calculated for each participant using the collected ECG data as input. RMSSD,

pNN50 and SDNN were calculated for each of the five recorded phases. Descriptive statistics for each of the three features for each of the five phases (“Baseline”, “Stroop introduction”, “Stroop test”, “Post questions”, and “Cooldown”) were calculated. Further, trajectories of the values for each participant were plotted on a line graph and visually inspected.

To examine associations between physiological and self-reported stress responses, participant-level change scores were calculated for each HRV feature (RMSSD, SDNN, and pNN50). For each participant, the value recorded during the “Baseline” phase was paired separately with the values recorded during the “Stroop test” and “Cooldown” phases. Changes were calculated as the target-phase value minus the baseline value. In addition, relative changes were calculated as $100 \times (\text{target phase} - \text{baseline}) / \text{baseline}$. Participants without valid values for both the baseline and respective target phase were excluded from the corresponding analysis. For pNN50, cases with a baseline value of zero were excluded from relative-change analyses to avoid undefined percentage changes.

Self-reported stress change was calculated from the single-stress-question responses as the post-Stroop rating minus the pre-Stroop rating, with positive values indicating an increase in perceived stress. Separate ordinary least-squares linear regressions were then fitted for each HRV feature and target phase, resulting in six models in total: RMSSD, SDNN, and pNN50 changes from baseline to “Stroop test”, and changes from baseline to “Cooldown”. In each model, the relative HRV change served as the predictor variable and the change in self-reported stress served as the outcome variable. Scatterplots including participant-level observations and fitted regression lines were generated for visual inspection. The regression coefficients, Pearson correlation coefficients, coefficients of determination (R^2), standard errors, and unadjusted p -values were retained to descriptively compare associations obtained for the “Stroop test” and “Cooldown” phases.

Head movement evaluation

An exploratory classification analysis was conducted to investigate whether head-movement data could distinguish between the “Baseline” and “Cooldown” phases. These phases used the same participant prompt but were separated by the Stroop stressor. Consequently, differences in movement patterns between the phases could provide an initial indication of whether head-movement data may be sensitive to changes associated with the experimental procedure. For each participant, the recorded data were separated into individual files for the “Baseline” and “Cooldown” phases and resampled to a uniform frequency of 100 Hz. As the source data were recorded at irregular timestamp intervals, values at the discrete resampling time points were interpolated. Rotational data represented as quaternions were interpolated using spherical linear interpolation (SLERP). The resampled recordings were then divided into non-overlapping 10-second intervals, resulting in labelled samples containing 1,000 time steps each. The resulting dataset was split into training, validation, and

test sets with proportions of 70%, 15%, and 15%, respectively. To prevent data leakage, all samples originating from the same recording were assigned exclusively to one of these sets.

Before model training, Uniform Manifold Approximation and Projection (UMAP) (McInnes et al., 2018) and t-distributed Stochastic Neighbor Embedding (t-SNE) (Van der Maaten & Hinton, 2008) were applied to visually inspect the distribution and potential separability of samples from both phases. Subsequently, four supervised classifiers were trained using scikit-learn (Pedregosa et al., 2011): a support vector machine (SVM) with a radial basis function (RBF) kernel, logistic regression, a k -nearest-neighbours (kNN) classifier, and a multi-layer perceptron (MLP) classifier. For logistic regression, the regularization parameter C was evaluated using the values 0.00001, 0.001, 0.01, 0.1, 1.0, 10.0, and 100.0. For the kNN classifier, the number of neighbours was evaluated using 1, 3, 5, 7, 11, 15, and 21. The MLP classifier consisted of one hidden layer with 32 units, used the ReLU activation function and Adam optimizer, and was configured with an L_2 regularization parameter of 0.0001, an initial learning rate of 0.001, and a batch size of 32.

For model interpretation, feature importance was examined for the logistic regression classifier based on its fitted model coefficients. For the kNN and SVM classifiers, trajectory permutation importance was calculated. In this procedure, individual input trajectories were permuted and the resulting change in classification performance was measured. The permutation procedure was repeated 30 times for each trajectory to obtain a more robust estimate of its importance.

5.1.3 Population and recruitment

Participants were recruited through direct intercept recruitment of students and employees on the campus of USTP as well as on social media channels used by USTP's students. Additionally, an e-mail was sent to students of the study program "Gesundheits- und Krankenpflege^{PLUS}". This was done as students in nursing adjacent fields were deemed desirable participants for this study as the library is aimed to be used in studies on nurses in the future.

In total 14 participants were recruited for this part of this thesis. Seven male and seven female participants, ranging from 20 to 36 years, were recruited, with a median age of 25. Three of the participants were active students of "Gesundheits- und Krankenpflege^{PLUS}"

5.2 Take-home evaluation

In addition to the supervised on-site evaluation, a small exploratory take-home evaluation was conducted. Its primary purpose was to assess whether participants could independently operate the developed system and repeatedly collect multimodal data outside a supervised laboratory environment. The evaluation also provided repeated measurements for exploring within-participant patterns in the recorded HRV features. Due to the small sample size, the take-home evaluation was treated as a feasibility evaluation rather than a formal assessment of reliability or clinical validity.

5.2.1 Study design

Participants were asked to use the application once per day on seven consecutive days, resulting in seven expected recording sessions per participant. They were permitted to retain the equipment for longer than one week to provide flexibility for equipment handover and return. However, the intended recording period for each participant remained one consecutive week.

Each participant received a Meta Quest 3 VR headset, a Polar H10 heart rate monitor, a headset charger, a carrying case, and spare batteries for the headset controllers. The application was installed and configured before the equipment was handed over. Participants additionally received printed instructions explaining how to put on the Polar H10, connect it to the headset, start the application, and conduct a recording session. These instructions are included in Appendix B. A telephone number and an email address were provided in case assistance was required during the evaluation.

The use of the equipment and the recording procedure were demonstrated during the handover. One participant completed their first recording session under supervision using the take-home version of the application. This session followed the same procedure as the subsequent take-home sessions and was therefore included in the analysis. The other participant began the recording period without completing a supervised practice session.

The take-home version used the same five phases and phase durations as the application used in the on-site evaluation: “Baseline”, “Stroop introduction”, “Stroop test”, “Post questions”, and “Cooldown”. A complete daily session lasted approximately eight minutes. Participants were not required to perform the recording at a particular time of day, and no restrictions concerning the recording location, food intake, exercise, or other daily activities were specified.

The numerical stress questions used in the on-site evaluation were retained in the take-home version of the application. Participants rated their perceived stress before and after

the Stroop test using the same response scale as in the on-site evaluation. These ratings allowed changes in self-reported stress to be calculated analogously to the on-site analysis.

The verbal usability question used during the on-site evaluation was replaced with a question intended to collect contextual information about events that might have influenced the participant's stress level on the respective day. During the "Post questions" phase, participants were asked:

"Did you have any events that might impact today's stress level? Do you feel especially relaxed today?"

Participants answered the question verbally, and their responses were stored as audio recordings on the headset. The verbal responses provided contextual information in addition to the numerical stress ratings.

5.2.2 Recorded data and data storage

The application recorded the same modalities and used the same sampling procedures as in the on-site evaluation. Head-movement data, heart-rate data, raw ECG data, audio recordings, timestamps, and phase information were stored in separate files where applicable. Heart-rate and raw ECG data were stored separately. RR intervals were not recorded as a separate signal but were derived from the raw ECG by the HRV analysis pipeline.

During the take-home period, all recordings were stored locally on the Meta Quest 3. The data were not transmitted to an external device or online service while the equipment was in the participants' homes. Each participant and each recording were identified using a pseudonymous UUID version 4 identifier. Names and other directly identifying information were not included in the recorded filenames.

After the equipment was returned, the recording archives were transferred from the headset using SideQuest. The transferred files were stored in an encrypted DataVault on the author's network-attached storage system. After the successful transfer of the recordings had been verified, the data were deleted from the headsets.

The mapping required to associate the pseudonymous identifiers with the participants was stored separately by the author. It was retained exclusively for fulfilling data-subject rights under the GDPR. Pseudonymized study data could be shared with other researchers, while the mapping keys remained with the author.

5.2.3 Evaluation procedure

After the equipment was returned, the transferred recording folders were manually inspected by the author. For each expected recording session, the available files and represented application phases were checked. A session was classified as complete when all five expected phases and the corresponding head-movement, heart-rate, raw ECG, and question-response recordings were present. A session was classified as partial when at least one expected recording file was present but the requirements for a complete session were not fulfilled.

This classification was used to describe the completeness of the data collection process. No numerical adherence score was calculated. Instead, the numbers of expected, complete, partial, and missing sessions were summarized descriptively. Because the stored files did not always permit the cause of an incomplete recording to be reconstructed, incomplete sessions were categorized based on the available data rather than an assumed technical cause.

Head-movement recordings were inspected for completeness but were not included in the classification analysis described for the on-site evaluation. All available audio recordings were manually transcribed verbatim by the author. The transcripts were examined descriptively to summarize events that participants considered potentially relevant to their stress level. No formal qualitative coding or thematic-analysis procedure was applied, and the transcriptions and summaries were not independently verified by a second researcher.

5.2.4 HRV and self-reported stress analysis

The raw ECG recordings were processed using the same in-house analysis pipeline as for the on-site evaluation (Iselor, 2026). RR intervals were derived from the raw ECG, after which RMSSD, SDNN, and pNN50 were calculated separately for the five application phases. Recordings or individual phase comparisons for which the required ECG data were incomplete or unavailable were excluded from the corresponding HRV analysis. A missing HRV estimate for the “Stroop introduction” phase did not result in exclusion from baseline-relative analyses when valid data were available for the “Baseline” and respective target phases.

The extracted HRV features were analyzed separately for each participant and recording day. Descriptive tables and plots were generated to examine changes across the application phases and the seven consecutive recording days. The numerical stress ratings collected before and after the Stroop test were processed using the same procedure as in the on-site evaluation. The change in self-reported stress was calculated by subtracting the pre-Stroop rating from the post-Stroop rating.

Where valid values for the required phases were available, absolute and relative changes from the “Baseline” phase to the “Stroop test” and “Cooldown” phases were calculated analogously to the on-site evaluation. Relative changes were calculated as

$$100 \times \frac{\text{target phase} - \text{baseline}}{\text{baseline}}.$$

A phase comparison was omitted when no valid value was available for either the baseline or the corresponding target phase. Relative changes in pNN50 were not calculated when the baseline pNN50 value was zero.

Where the required ECG and numerical stress data were available, the relationship between changes in HRV and changes in self-reported stress was examined using the same exploratory analysis procedure as for the on-site evaluation. However, the take-home observations were interpreted separately for each participant because only two participants took part and repeated daily measurements from the same participant cannot be considered independent observations. The analysis was therefore descriptive and exploratory rather than a population-level assessment of statistical reliability or validity.

The transcribed verbal responses were used to contextualize the daily numerical stress ratings and HRV measurements. No formal qualitative coding or thematic-analysis procedure was applied.

5.2.5 Population and recruitment

Two adult students from the University of Applied Sciences St. Pölten were recruited for the take-home evaluation through direct referral, constituting a convenience sample. One participant was enrolled in the “Gesundheits- und Krankenpflege^{PLUS}” programme. One of the two participants had previously taken part in the on-site evaluation, whereas the other had not. Participation was voluntary, and no compensation was provided.

6 Results

This chapter presents the results of both the on-site evaluation and the take-home evaluation of the application and library. The respective sections show the results achieved during the phases.

6.1 On-site evaluation

This section includes the results of the On-site evaluation of the application described in chapter 4 as described in section 5.1 of chapter 5.

6.1.1 SUS scores

The System Usability Scale (SUS) results from the on-site evaluations were consistently high. Scores ranged from 75 to 100, with a mean of 85. Figure 6.1 illustrates the distribution of the obtained scores.

Figure 6.2 depicts the mean response for each SUS item. Responses to negatively worded items were reverse scored before averaging. The lowest mean item score was observed for Q1, “I think that I would like to use this system frequently”, with a value of 3.07. The highest mean item score was observed for Q5, “I found the various functions in this system were well integrated”, with a value of 4.64.

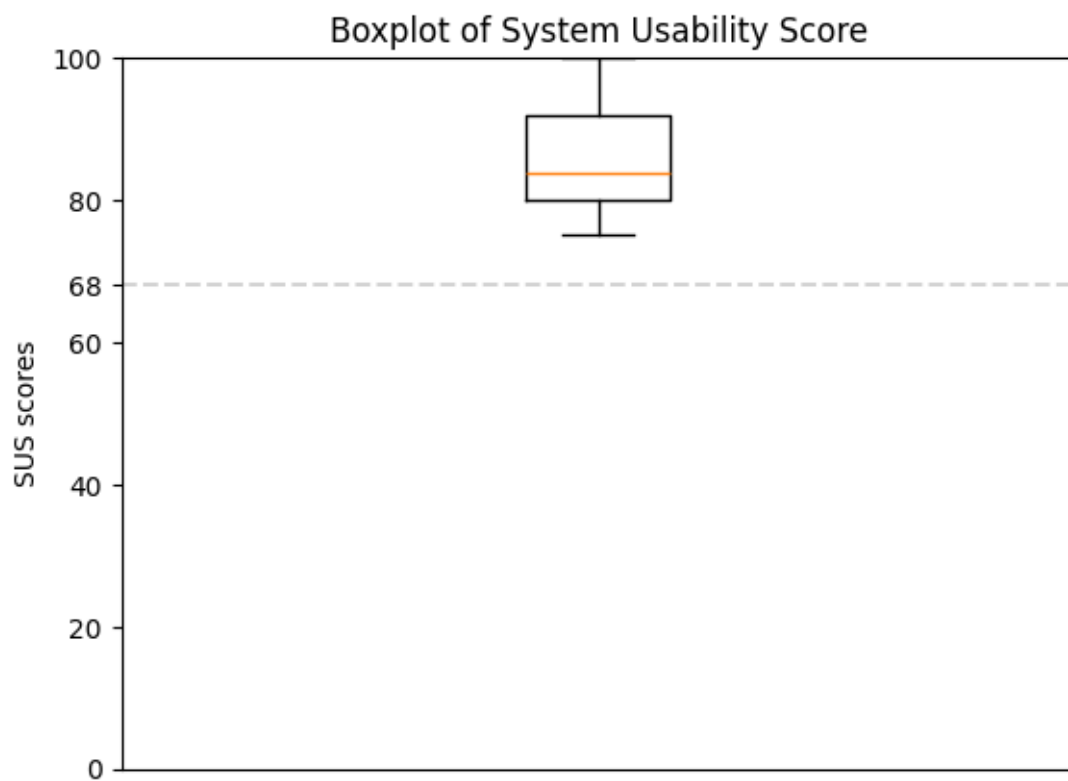


Figure 6.1. Boxplot showing the results of the SUS administered during the on-site evaluation. Additionally a light gray dotted line indicates a SUS score of 68.

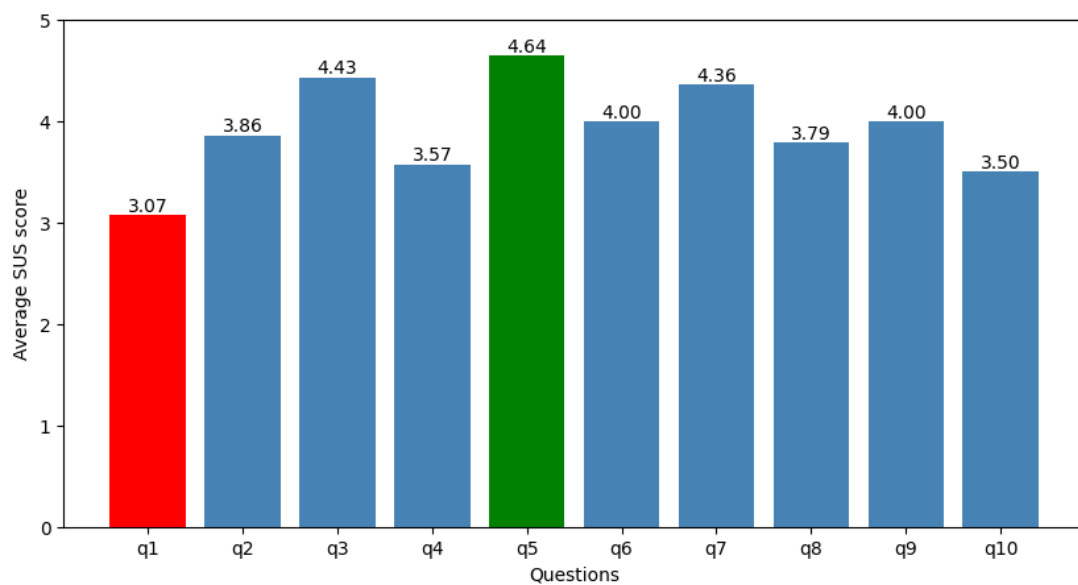


Figure 6.2. Bar chart depicting the average answer to the SUS questions. The values for the odd questions are already inverted. The highest scoring question (q5) is highlighted in green. The lowest scoring question (q1) is highlighted in red.

6.1.2 Analysis of verbal feedback

All 14 participants of the on-site evaluation managed to record an audio file containing verbal feedback to the application. All participants responded in either German or English. Six participants mentioned, that they enjoyed the Stroop Room test as a game. A number of participants said, that they liked the 360° nature of the application, although it also was one of the most frequently mentioned difficulties in answering the Stroop challenges correctly. The most commonly mentioned issue was initial confusion about the Stroop test's rules. Three participants explicitly mentioned that the Stroop test was stressful, compared to a single participant who mentioned that the test was not overly stressful to them. Seven of the 14 voice recordings suggested that participants experienced the Stroop effect mentioning mix-ups and problems with the color word test. One of the participants mentioned that they adapted strategy mid-test and opted to not turn their head looking at the colored walls but instead chose to mentally remember the room layout and only point the hands in the desired direction. That participant mentioned, that this particular strategy reduced the perceived stress for them. Three participants mentioned problems differentiating the used yellow, orange and red.

None of the verbal feedback contained any mention of problems with the application's user interface or with the connection of the heart rate monitor to the headset.

6.1.3 Single-Question Likert scale questionnaire

As described in chapter 5, participants answered "How stressed do you feel?", on a five point Likert scale, before and after the Stroop Room test. The answers before the Stroop test ranged between one and two with a mean value of 1.29. After the Stroop test answers ranged between two and four with a mean value of 3.07. The mean difference between the two measurement was 1.78. A box plot illustrating those results can be seen in figure 6.3.

All 14 participants reported a higher stress rating after the Stroop Room test than before the test.

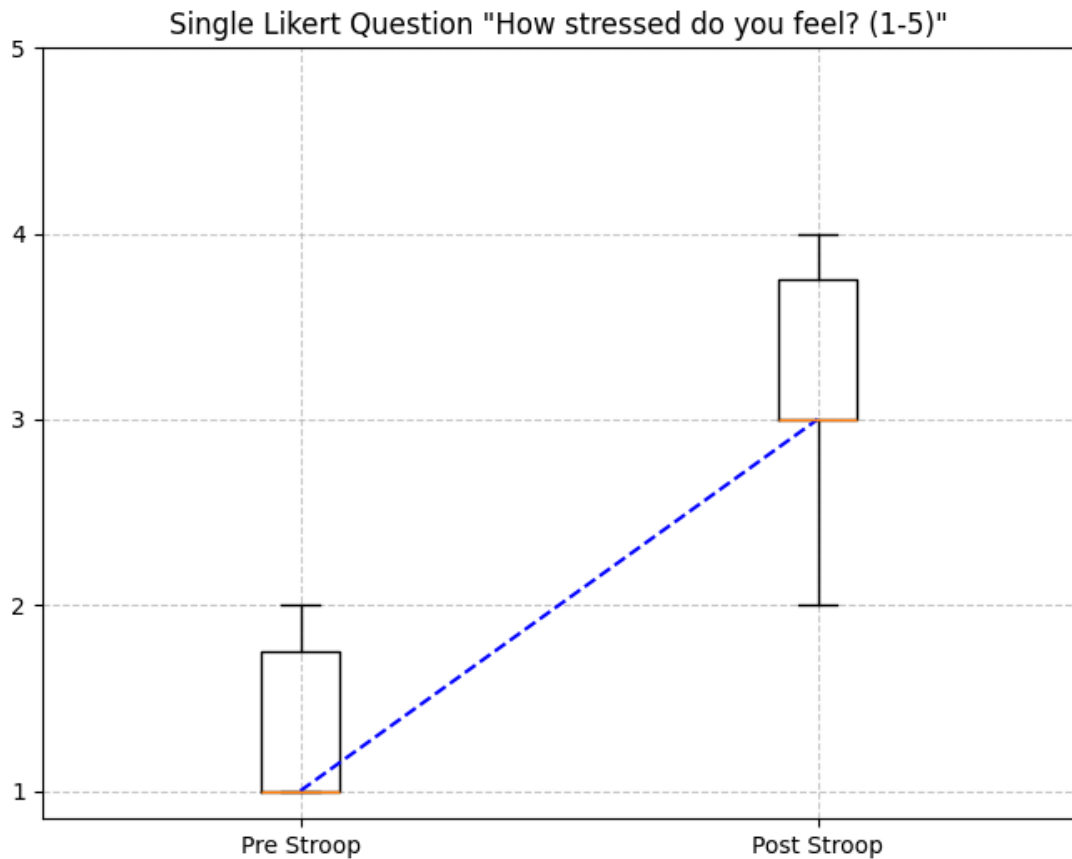


Figure 6.3. Boxplot depicting the answers to the question “How stressed do you feel?”, before and after the Stroop Room test was performed.

6.1.4 Heart Rate Variability features

Valid ECG-derived HRV values were available for 13 of the 14 participants. Therefore, all HRV analyses were based on $n = 13$. RMSSD, SDNN, and pNN50 were calculated for each participant during each of the five task phases. RMSSD and SDNN are reported in milliseconds, whereas pNN50 is reported as a percentage. The boxplots illustrate the distributions across participants, while the trajectory plots show within-participant changes between phases.

RMSSD

Descriptive statistics for participants’ RMSSD are presented in Table 6.1. The highest mean RMSSD was observed during baseline (57.80 ms), followed by the Stroop test phase (54.48 ms). Mean RMSSD decreased during the introduction phase (46.70 ms), rose again during testing and reached its lowest value during the post question phase (34.23 ms). Mean RMSSD increased to 45.76 ms during cooldown, while remaining below the baseline level.

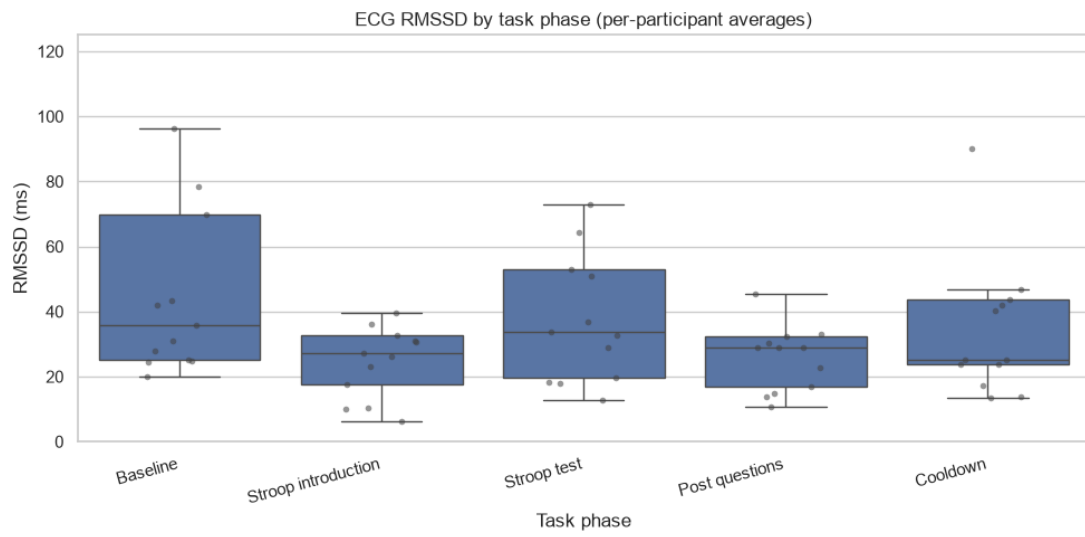


Figure 6.4. Boxplot of the RMSSD by task phase of all 13 recordings

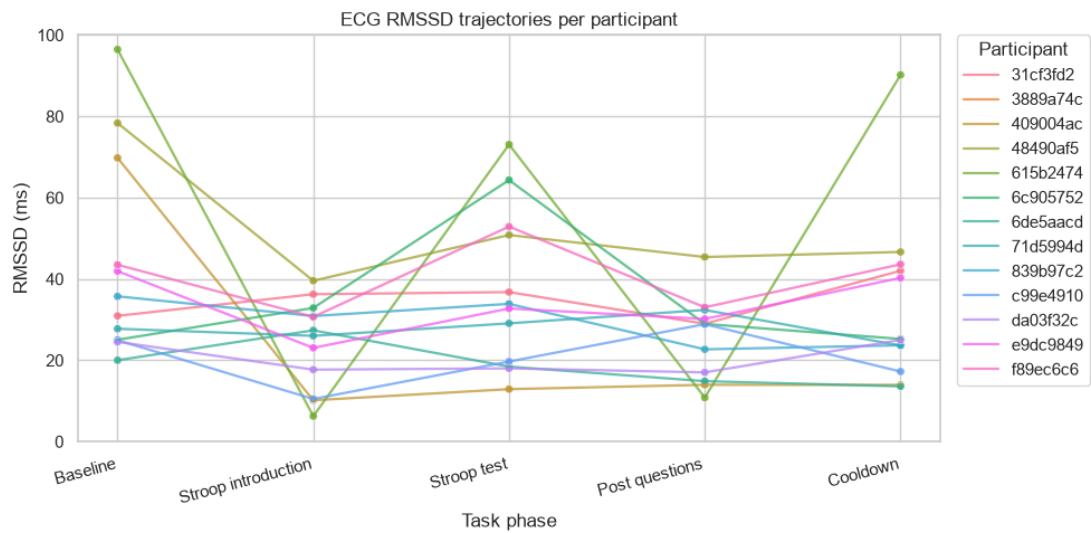


Figure 6.5. Trajectories of the RMSSD for all 13 recordings during testing

Task Phase	<i>n</i>	Mean	Median	SD	Min	Max
Baseline	13	57.80	35.73	57.57	20.02	232.33
Stroop introduction	13	46.70	27.36	81.52	6.26	315.78
Stroop test	13	54.48	33.88	66.12	12.90	265.63
Post questions	13	34.23	28.95	32.57	10.82	137.84
Cooldown	13	45.76	25.25	47.83	13.60	189.78

Table 6.1. Descriptive statistics for RMSSD across the five task phases. Values are rounded to two decimal places.

For RMSSD the mean was higher than the median every phase showing the largest difference during the introduction phase. During introduction mean RMSSD was 46.70 ms and the median was 27.36 ms. Together with the large standard deviation of 81.52 ms and a maximum value of 315.78 ms, this indicates that the mean for this phase was strongly influenced by comparatively high individual values. Considerable dispersion was also present during baseline and Stroop test phases. The smallest difference between mean and median was present during the post question phase, which also had the lowest standard deviation (32.57 ms).

The RMSSD distribution across phases is shown in Figure 6.4. The trajectories shown in Figure 6.5 show that changes in RMSSD are highly individual between participants. While several participants exhibited lower values during and after the Stroop test, the magnitude of this reduction varied greatly. For some of the participants a trajectory pattern in the shape of a W is visible.

SDNN

Task Phase	<i>n</i>	Mean	Median	SD	Min	Max
Baseline	13	64.69	55.28	28.06	38.37	146.59
Stroop introduction	13	50.77	45.43	43.39	13.33	185.39
Stroop test	13	62.90	57.37	31.58	33.82	156.58
Post questions	13	55.42	55.82	22.48	25.78	109.64
Cooldown	13	58.41	52.96	23.92	27.74	120.25

Table 6.2. Descriptive statistics for SDNN values across the five task phases. Values are rounded to two decimal places.

Table 6.2 presents the descriptive SDNN statistics. Mean SDNN was highest during the baseline phase (64.69 ms) and lowest during the introduction phase (50.77 ms). It increased to 62.90 ms during the Stroop test and decreased back to 55.42 ms during the post question phase. During cooldown mean SDNN increased slightly to 58.41 ms. Mean SDNN remained below baseline during all subsequent phases.

The medians followed a generally similar pattern to the mean, although the highest median was observed during the Stroop test (57.37 ms) rather than during the baseline phase

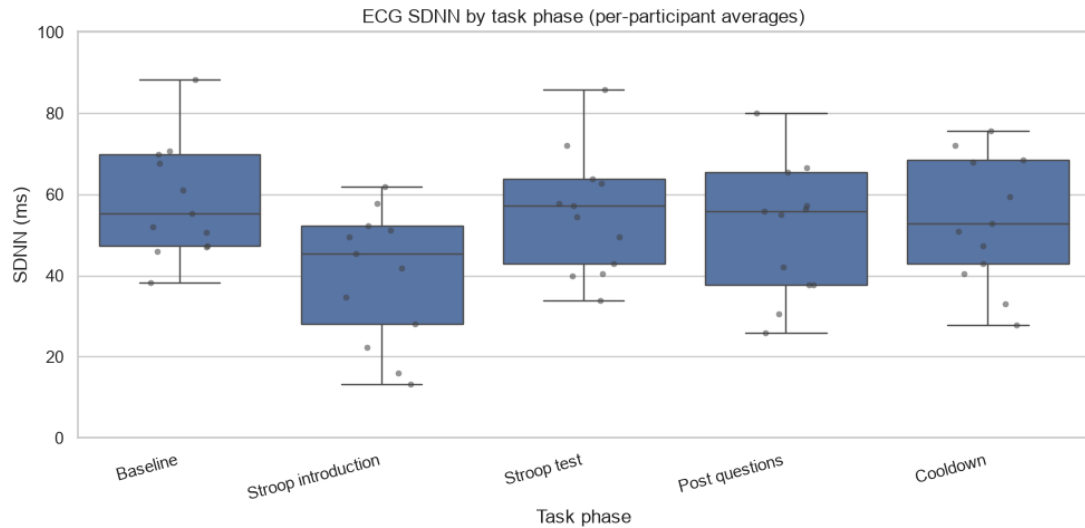


Figure 6.6. Boxplot of the SDNN by task phase of all 13 recordings

(55.28 ms). The greatest standard deviation of 43.39 ms and the widest observed range, from 13.33 to 185.39 ms was observed during the introduction phase. Variability was lowest during post question and cooldown phases.

The SDNN distribution across phases is shown in Figure 6.6. As illustrated by the crossing lines in Figure 6.7, the phase-related changes differed greatly between participants. Although a reduction from baseline to the later phases was visible for several participants this pattern was not observed for every participant.

6 Results

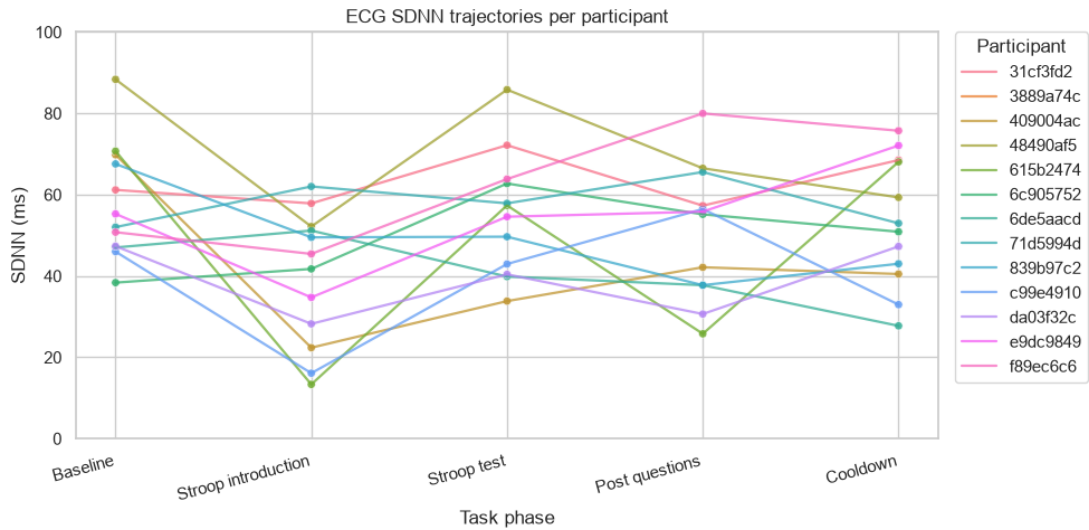


Figure 6.7. Trajectories of the SDNN for all 13 recordings during testing

pNN50

Task Phase	<i>n</i>	Mean	Median	SD	Min	Max
Baseline	13	11.54	6.58	10.64	1.82	36.46
Stroop introduction	13	9.23	6.48	10.24	0.00	35.71
Stroop test	13	8.81	8.14	7.82	0.63	27.35
Post questions	13	7.70	6.72	6.29	0.00	19.64
Cooldown	13	9.13	4.83	8.75	0.00	26.32

Table 6.3. Descriptive statistics of pNN50 values across the five task phases. Values are expressed as percentages and rounded to two decimal places.

Descriptive statistics for pNN50 can be seen in Table 6.3. Mean pNN50 was highest during baseline (11.54%) and decreased during both the introduction (9.23%) and Stroop test (8.81%) phases. The lowest mean of 7.70% was observed during the post question phase, followed by an increase to 9.13% during the cooldown phase. As with RMSSD and SDNN, the mean did not return to its baseline level during cooldown.

Unlike the mean, the highest median pNN50 was observed during the Stroop test phase (8.14%) with the lowest occurring during the cooldown phase (4.83%). Mean values were higher than median values in all phases, particularly during baseline and cooldown, indicating that comparatively high values from a subset of participants influenced the phase means.

Figure 6.8 shows the positively skewed pNN50 distributions and the presence of comparatively high participant values. The trajectories in Figure 6.9 further demonstrate marked interindividual differences. Some participants maintained relatively low pNN50 values across all phases, whereas others showed pronounced increases or decreases between individual

phases.

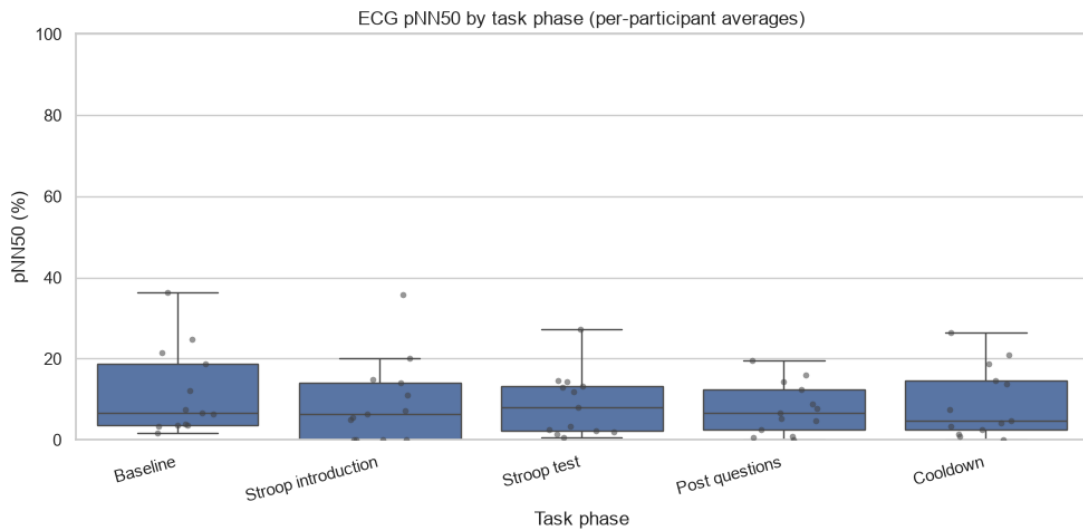


Figure 6.8. Boxplot of the pNN50 by task phase of all 13 recordings

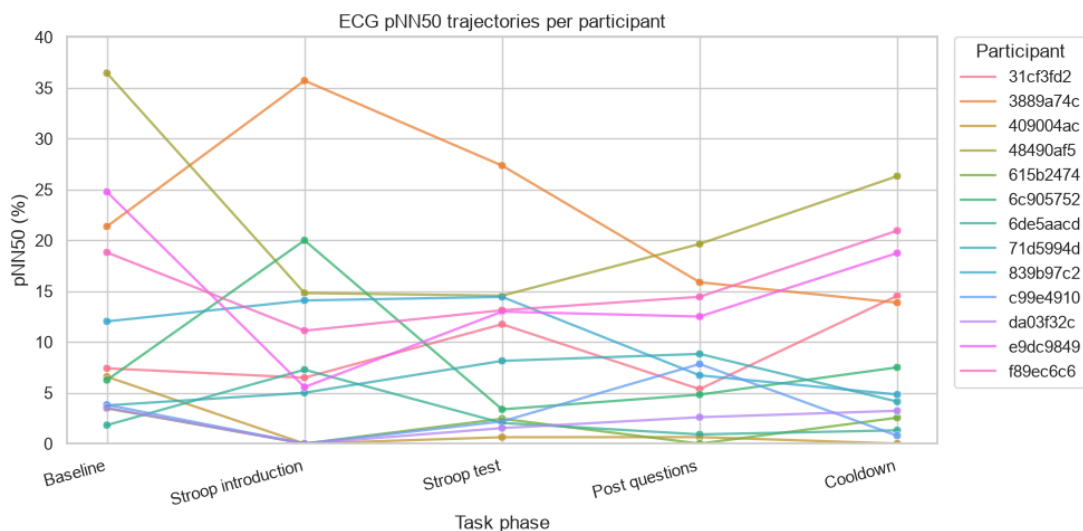


Figure 6.9. Trajectories of the pNN50 for all 13 recordings during testing

Summary of descriptive HRV patterns

At the group level, all three HRV features decreased in mean value from baseline to the Stroop introduction. Mean RMSSD and SDNN subsequently increased during the Stroop test, whereas mean pNN50 decreased slightly. All three means decreased during the post question phase and increased again during cooldown. Nevertheless, none of the mean values returned to its respective baseline level during cooldown.

The large standard deviations, differences between means and medians, and crossing participant trajectories indicate substantial interindividual variability. Consequently, the group

means do not represent a uniform response across participants. These results are descriptive and do not, by themselves, establish whether the observed differences between task phases were statistically significant

6.1.5 Association between HRV changes and self-reported stress change

The regression results for the associations between HRV change and self-reported stress change are presented in Table 6.4.

Table 6.4. Linear regression results for the association between baseline-relative HRV changes and changes in self-reported stress. Positive SSQ change scores indicate increased self-reported stress after the Stroop task.

Feature	Target phase	<i>n</i>	<i>r</i>	<i>R</i> ²	<i>p</i>	Slope
RMSSD	Stroop test	13	-0.295	0.087	.327	-0.0045
RMSSD	Cooldown	13	-0.141	0.020	.645	-0.0042
SDNN	Stroop test	13	-0.219	0.048	.473	-0.0065
SDNN	Cooldown	13	0.031	0.001	.919	0.0009
pNN50	Stroop test	13	-0.241	0.058	.428	-0.0035
pNN50	Cooldown	13	-0.249	0.062	.412	-0.0042

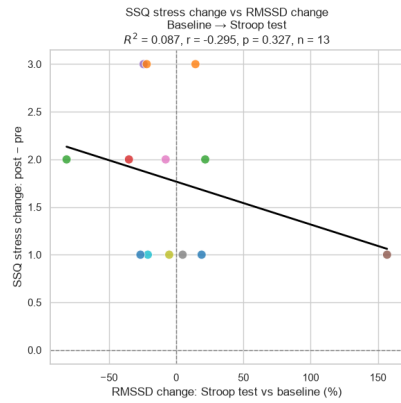
Note. HRV changes were calculated as relative percentage changes from baseline. *r* denotes the Pearson correlation coefficient. The slope represents the predicted change in self-reported stress per one percentage-point increase in the respective HRV feature. All reported *p*-values are unadjusted.

The corresponding participant-level scatterplots and fitted regression lines are shown in Figure 6.10.

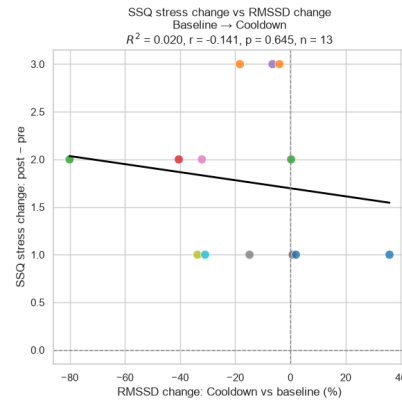
All regression models showed weak or negligible associations between baseline-relative HRV changes and changes in self-reported stress (Table 6.4). None of the six models reached statistical significance at the conventional threshold of $p < .05$. The unadjusted *p*-values ranged from $p = .327$ to $p = .919$. Furthermore, the models explained only a small proportion of the variance in self-reported stress change, with R^2 values ranging from .001 to .087. Thus, the regression analyses did not provide statistical evidence for a linear association between changes in the assessed HRV features and changes in self-reported stress in this sample.

For the baseline-to-“Stroop test” comparison, all three HRV features showed negative correlation coefficients with self-reported stress change. The largest observed association was found for RMSSD ($r = -0.295$, $R^2 = .087$, $p = .327$), followed by pNN50 ($r = -0.241$, $R^2 = .058$, $p = .428$) and SDNN ($r = -0.219$, $R^2 = .048$, $p = .473$). None of these associations was statistically significant.

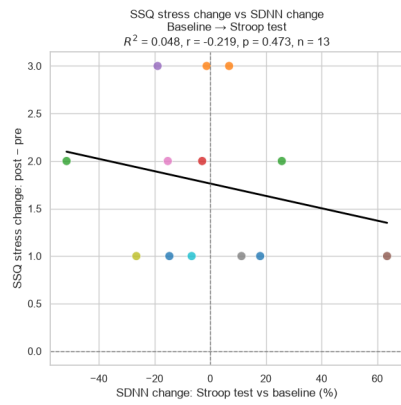
For the baseline-to-“Cooldown” comparison, RMSSD and pNN50 showed weak negative associations with self-reported stress change (RMSSD: $r = -0.141$, $R^2 = .020$, $p =$



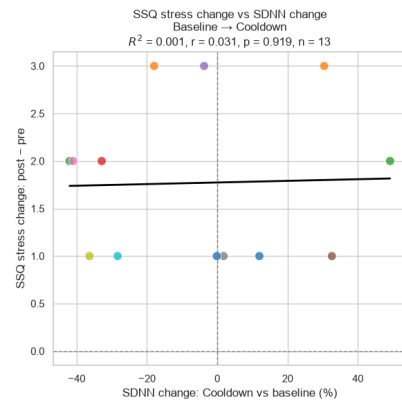
(a) RMSSD: Baseline to Stroop test



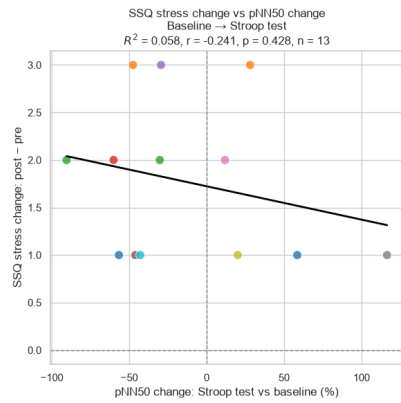
(b) RMSSD: Baseline to Cooldown



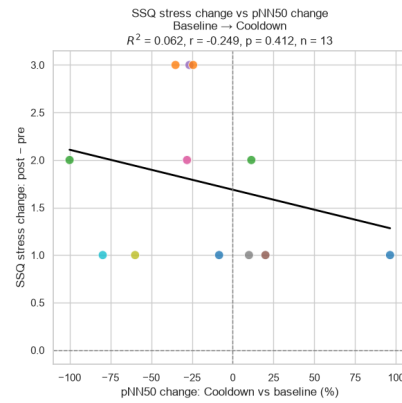
(c) SDNN: Baseline to Stroop test



(d) SDNN: Baseline to Cooldown



(e) pNN50: Baseline to Stroop test



(f) pNN50: Baseline to Cooldown

Figure 6.10. Associations between baseline-relative HRV changes and changes in self-reported stress. The left column shows changes from baseline to the “Stroop test” phase, whereas the right column shows changes from baseline to the “Cooldown” phase. Each point represents one participant ($n = 13$); black lines represent fitted linear regression lines.

.645; pNN50: $r = -0.249$, $R^2 = .062$, $p = .412$). SDNN showed an approximately zero association during cooldown ($r = 0.031$, $R^2 = .001$, $p = .919$). None of the cooldown-based models was statistically significant.

Comparing the two target phases descriptively, no consistent pattern was observed across the three HRV features. The baseline-to-“Stroop test” models showed larger R^2 values for RMSSD and SDNN, whereas the baseline-to-“Cooldown” model showed a slightly larger R^2 value for pNN50. However, all explained-variance values were small and none of the models was statistically significant. Therefore, the analyses did not identify either the “Stroop test” or “Cooldown” phase as consistently more strongly associated with self-reported stress change.

6.1.6 Head movement classification

As described in Chapter 5 the distribution of head-movement samples for the “Baseline” and “Cooldown” phases was examined using UMAP and t-SNE. Figure 6.11 shows the resulting two-dimensional projections.

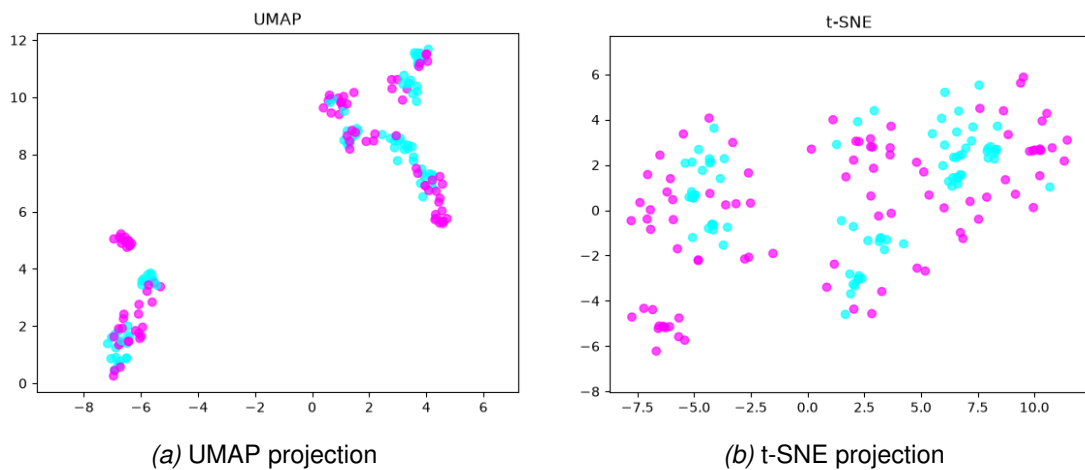


Figure 6.11. Two-dimensional projections of the head-movement samples from the “Baseline” and “Cooldown” phases. Each point represents one 10-second sample and is colored according to its phase. “Baseline” samples are colored cyan, “Cooldown” samples are colored magenta

In the UMAP projection, the samples formed several spatially separated groups. Samples from both the “Baseline” and “Cooldown” phases were present within most of these groups, although their relative proportions varied. The t-SNE projection showed a more dispersed distribution with several local groupings. The two phases overlapped across most regions of the projection, while some smaller regions contained a greater concentration of samples from one phase.

After segmentation, the dataset contained 257 ten-second samples, comprising 131 samples from the “Baseline” phase and 126 samples from the “Cooldown” phase. The training set contained 174 samples, the validation set contained 37 samples, and the test set contained 46 samples.

Hyperparameter selection

Hyperparameters for logistic regression and kNN were selected according to balanced accuracy on the validation set. Logistic regression achieved its highest validation balanced accuracy of .618 with $C = 0.001$. The balanced accuracies for the remaining values of C ranged from .461 to .617. Accordingly, $C = 0.001$ was selected for the final model.

For kNN, the highest validation balanced accuracy was .645 with $k = 1$. Across the remaining evaluated values of k , balanced accuracy ranged from .556 to .643. Consequently, $k = 1$ was selected for the final model.

Classification performance

The performance of the four classifiers on the held-out test set is presented in Table 6.5. The test set contained 46 samples, comprising 28 samples from class 0 and 18 samples from class 1.

Table 6.5. Performance of the head-movement classifiers on the held-out test set. Precision, recall, and F_1 -score are macro-averaged across the two classes.

Classifier	Accuracy	Balanced accuracy	Precision	Recall	F_1 -score
SVM	0.609	0.609	0.605	0.609	0.601
Logistic regression	0.674	0.643	0.655	0.643	0.647
kNN	0.783	0.732	0.825	0.732	0.743
MLP	0.630	0.607	0.610	0.607	0.607

Among the four classifiers, kNN achieved the highest test accuracy (0.783) and balanced accuracy (0.732). It also obtained the highest macro-averaged precision (0.825) and F_1 -score (0.743). Logistic regression achieved an accuracy of 0.674 and a balanced accuracy of 0.643. The SVM and MLP achieved balanced accuracies of 0.609 and 0.607, respectively.

For kNN, recall was .964 for “Baseline” and .500 for class “Cooldown”. Its corresponding class-specific F_1 -scores were .844 and .643. Logistic regression achieved recall values of .786 for “Baseline” and .500 for “Cooldown”. The SVM obtained recall values of .607 and .611, while the MLP obtained recall values of .714 and .500 for “Baseline” and “Cooldown”, respectively.

The confusion matrices for the four classifiers are shown in Figure 6.12. The SVM correctly classified 17 of the 28 “Baseline” samples and 11 of the 18 “Cooldown” samples. Logistic regression correctly classified 22 “Baseline” samples and nine “Cooldown” samples. The

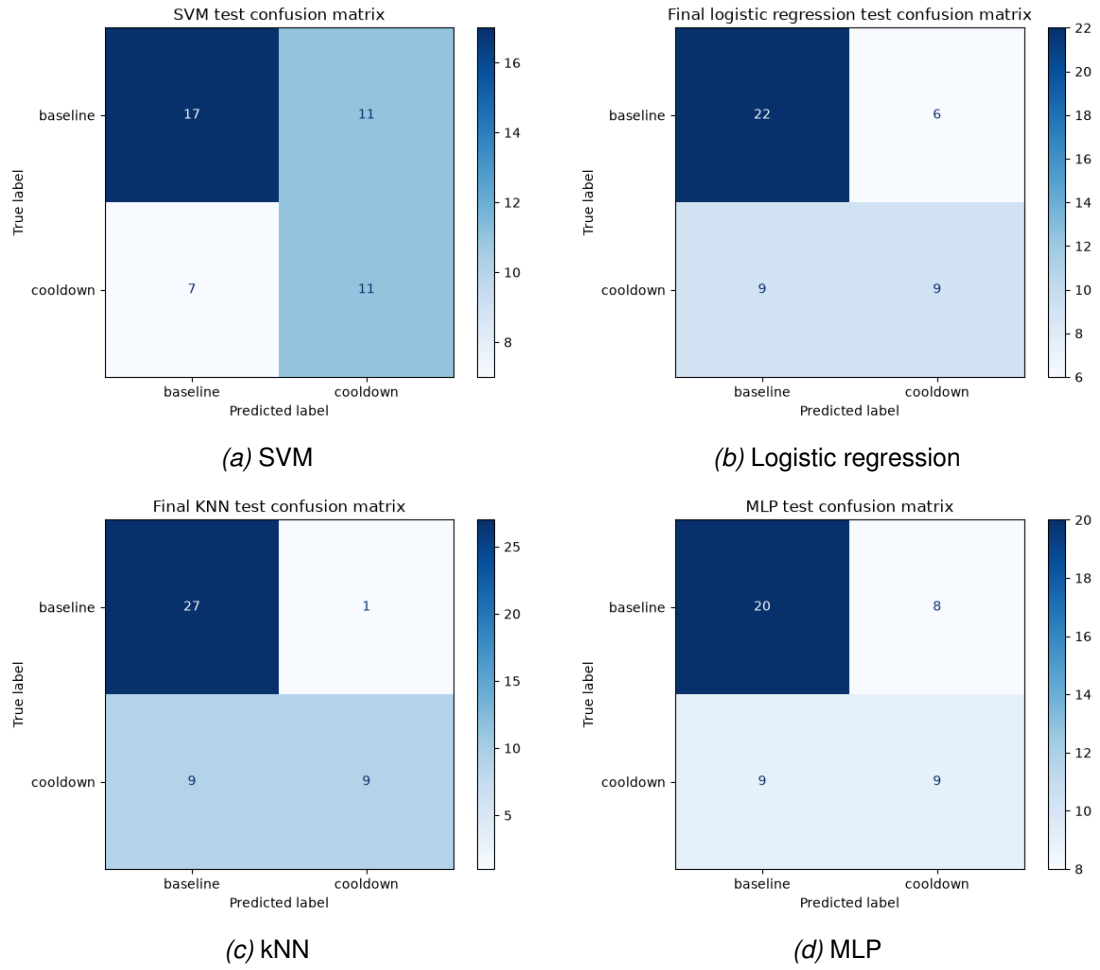


Figure 6.12. Confusion matrices for the four classifiers on the held-out test set. Rows represent the true phase labels and columns represent the predicted phase labels.

kNN classifier correctly classified 27 “Baseline” samples and nine “Cooldown” samples, while the MLP correctly classified 20 and nine samples, respectively.

The kNN classifier produced one false “Cooldown” prediction for the 28 “Baseline” samples. However, nine of the 18 “Cooldown” samples were classified as “Baseline”. Logistic regression and the MLP also correctly classified nine “Cooldown” samples, while the SVM correctly classified 11.

Feature importance

Feature importance was examined using the mean absolute coefficients of the logistic regression model and trajectory permutation importance for the SVM and kNN classifiers. The results are presented in Table 6.6.

Table 6.6. Feature-importance results for the head-movement classifiers. Logistic-regression importance is reported as the mean absolute coefficient. Permutation importance for SVM and kNN is reported as the mean decrease in balanced accuracy, with the standard deviation across 30 repetitions in parentheses.

Trajectory	Logistic regression	SVM	kNN
PosX	.00065	.060 (.056)	.201 (.051)
PosY	.00038	.000 (.000)	.010 (.022)
PosZ	.00098	.014 (.026)	.153 (.036)
RotX	.00092	-.035 (.023)	-.007 (.025)
RotY	.00078	.009 (.031)	.017 (.025)
RotZ	.00071	.003 (.031)	-.005 (.017)
RotW	.00030	-.013 (.013)	-.011 (.009)
VelX	.00139	.022 (.019)	.021 (.011)
VelY	.00115	.002 (.007)	.001 (.003)
VelZ	.00152	.029 (.027)	.023 (.013)
AngVelX	.00226	.007 (.012)	.002 (.006)
AngVelY	.00175	.009 (.014)	.017 (.015)
AngVelZ	.00161	.012 (.013)	-.020 (.014)

Note. Logistic-regression coefficients and permutation-importance values represent different measures and should not be compared numerically. Negative permutation-importance values indicate that balanced accuracy increased on average when the respective trajectory was permuted.

For logistic regression, AngVelX had the largest mean absolute coefficient (.00226), followed by AngVelY (.00175) and AngVelZ (.00161). The smallest mean absolute coefficients were obtained for RotW (.00030), PosY (.00038), and PosX (.00065).

For the SVM, permutation of PosX produced the largest mean decrease in balanced accuracy (.060, $SD = .056$). The next-largest decreases were obtained for VelZ (.029, $SD = .027$) and VelX (.022, $SD = .019$). Negative mean changes were obtained for RotW (-.013) and RotX (-.035).

For kNN, the largest mean decreases in balanced accuracy occurred when PosX (.201,

$SD = .051$) and $PosZ (.153, SD = .036)$ were permuted. All remaining trajectories produced mean decreases below .024. Negative mean changes were obtained for $RotZ$, $RotX$, $RotW$, and $AngVelZ$.

6.2 Week-long take-home evaluation

This section presents the results of the two week-long take-home evaluations described in Section 5.2. The results concern the completeness of the independently collected recordings, technical observations, self-reported stress, verbal responses, and the available ECG-derived HRV features.

6.2.1 Recording completeness and technical observations

Across both participants, 14 recording sessions were created. Participant 1 created eight recordings, whereas Participant 2 created six recordings. Although seven recordings were expected from each participant, Participant 1 completed one additional recording after encountering a tracking problem. Participant 2 did not complete one of the seven expected sessions.

Table 6.7 summarizes the completeness of the take-home recordings. All eight recordings created by Participant 1 were classified as complete. Of the six recordings created by Participant 2, three were complete and three were partial. Consequently, one expected recording from Participant 2 was missing.

Table 6.7. Completeness of the take-home recording sessions.

Participant	Expected	Recorded	Complete	Partial	Missing
Participant 1	7	8	8	0	0
Participant 2	7	6	3	3	1
Total	14	14	11	3	1

The partial recordings created by Participant 2 were missing raw ECG data required for the planned HRV analysis. Raw ECG was unavailable throughout recordings 4 and 6. In recording 5, raw ECG was available during the “Baseline” phase but not during the remaining phases. Heart-rate data were nevertheless recorded during all six sessions. Head-movement data were complete and usable in all recordings from both participants.

Neither participant contacted the author for technical assistance during the take-home evaluation. Both participants replaced the batteries of the Meta Quest 3 controllers once during the evaluation.

Participant 1 encountered a tracking failure during one recording. During this failure, the image displayed in the headset remained fixed at the virtual camera position and did not update according to the participant's head position. The application continued operating, and the participant completed and saved the affected session. The participant then repeated the session immediately afterwards, resulting in the eighth recording. The problem was additionally described by the participant in the verbal response recorded during the affected session.

6.2.2 Self-reported stress

In every recording, the participants rated their perceived stress on a five-point response scale before and after the Stroop test. Table 6.8 presents the individual ratings and corresponding change scores. Positive values indicate higher self-reported stress after the Stroop test, whereas negative values indicate lower self-reported stress.

Table 6.8. Pre- and post-Stroop self-reported stress ratings during the take-home evaluation. Change scores were calculated as the post-Stroop rating minus the pre-Stroop rating.

Participant	Recording	Pre-Stroop	Post-Stroop	Change
Participant 1	1	1	3	2
Participant 1	2	3	2	-1
Participant 1	3	2	3	1
Participant 1	4	2	4	2
Participant 1	5	2	2	0
Participant 1	6	3	3	0
Participant 1	7	3	3	0
Participant 1	8	1	1	0
Participant 2	1	2	4	2
Participant 2	2	3	3	0
Participant 2	3	2	3	1
Participant 2	4	2	3	1
Participant 2	5	4	4	0
Participant 2	6	2	2	0

Participant 1 had a mean pre-Stroop rating of 2.13 and a mean post-Stroop rating of 2.63. The resulting mean change was 0.50 points. Three of the eight recordings showed an increase in self-reported stress, four showed no change, and one showed a decrease of one point.

Participant 2 had a mean pre-Stroop rating of 2.50 and a mean post-Stroop rating of 3.17. The resulting mean change was 0.67 points. Three of the six recordings showed an increase in self-reported stress, and three showed no change. Participant 2 did not report a decrease in stress after the Stroop test.

For both participants, all recordings from recording 5 onward had a change score of zero. Thus, neither participant reported a difference between the pre- and post-Stroop ratings during these later recordings. This result is reported descriptively; possible explanations for the observed pattern are considered in the discussion.

6.2.3 Verbal responses

Both participants produced a verbal response during every recorded session. The complete transcripts are provided in Appendix D. The responses described contextual factors that the participants considered relevant to their stress on the respective day. As specified in the Method chapter, the responses were examined descriptively and were not subjected to formal qualitative coding.

Participant 1 primarily reported factors related to academic work, including assignments, studying, and examinations. Additional contextual factors included organizational responsibilities, high ambient temperatures, and the completion of the take-home evaluation. During one recording, Participant 1 reported that no particular stressor had occurred. In the session affected by the tracking failure, the participant used the verbal response to describe the technical problem and stated that the inability to perform the Stroop challenges as intended was distressing.

Participant 2 described days with both comparatively low and comparatively high perceived stress. Reported contextual factors included leisure activities, household tasks, poor sleep, exercise, organizational work, an upcoming film production, and an assessment completed earlier on the same day. In one recording, Participant 2 stated that their position within the virtual scene and the headset's low battery level were adding to their stress. In another recording, the Stroop test reminded the participant of a similar task completed during an earlier assessment, which the participant reported as increasing their perceived stress.

Overall, the recordings demonstrated that both participants were able to independently record verbal contextual information throughout the take-home evaluation. Most responses concerned events outside the application, while a small number also documented application- or equipment-related stressors.

6.2.4 ECG and HRV data availability

Participant 1 recorded raw ECG data during all eight sessions. Participant 2 recorded sufficient raw ECG data for the planned baseline-relative HRV comparisons during the first three sessions. During recording 4, raw ECG was not present. During recording 5, raw ECG was present only during the “Baseline” phase, and during recording 6 it was not

present.

Consequently, baseline-to-“Stroop test” and baseline-to-“Cooldown” HRV changes were available for all eight recordings from Participant 1 and for the first three recordings from Participant 2. The absence of raw ECG did not affect the availability of the numerical stress ratings or head-movement recordings.

HRV values for the “Stroop introduction” phase were unavailable in several otherwise complete sessions. This occurred when the participant proceeded through the already familiar instructions before a sufficiently long ECG segment had been recorded for HRV calculation. These missing phase-level estimates did not prevent the calculation of changes from “Baseline” to “Stroop test” or “Cooldown”.

6.2.5 Descriptive HRV results

RMSSD, SDNN, and pNN50 varied between phases and recording sessions for both participants. Due to the small number of participants and the limited number of valid sessions for Participant 2, the results were examined separately for each participant.

For Participant 1, baseline RMSSD values ranged from 18.86 to 54.16 ms, baseline SDNN values ranged from 45.11 to 122.88 ms, and baseline pNN50 values ranged from 1.99% to 36.27%. The range of baseline values and the phase-level trajectories indicate substantial variation between recording sessions.

Participant 1 showed a higher RMSSD during the “Stroop test” than during “Baseline” in five of the eight recordings and a lower value in three recordings. In contrast, RMSSD during “Cooldown” was lower than during “Baseline” in six recordings and higher in two.

SDNN during the “Stroop test” was lower than during “Baseline” in six of the eight recordings and higher in two. The same counts were observed for the comparison between “Baseline” and “Cooldown”. Participant 1’s pNN50 was lower during the “Stroop test” than during “Baseline” in five recordings and higher in three. For the “Cooldown” comparison, pNN50 was lower than baseline in four recordings and higher in four.

For Participant 2, baseline-to-target comparisons were available for the first three recordings. RMSSD and SDNN were lower during the “Stroop test” than during “Baseline” in recordings 1 and 2, but higher in recording 3. During “Cooldown”, both features were lower than baseline in recording 1 and higher in recordings 2 and 3.

For Participant 2, pNN50 decreased from “Baseline” to “Stroop test” in recordings 1 and 2 and increased in absolute terms in recording 3. Similarly, pNN50 decreased from “Baseline” to “Cooldown” in recording 1 and increased in recordings 2 and 3. Because baseline pNN50 was zero in recording 3, relative pNN50 changes could not be calculated for that

recording.

The direction and magnitude of HRV changes therefore differed across recordings for both participants. No phase-related change was consistently observed across all valid sessions and all three HRV features.

6.2.6 Association between HRV and self-reported stress changes

Participant-specific exploratory regressions were calculated for Participant 1 using the eight available recordings. Table 6.9 presents the associations between baseline-relative HRV changes and changes in self-reported stress. The small number of observations means that the estimates are descriptive and should not be interpreted as reliable population-level estimates.

Table 6.9. Participant 1 regression results for the association between baseline-relative HRV changes and changes in self-reported stress during the take-home evaluation.

Feature	Target phase	<i>n</i>	<i>r</i>	R^2	<i>p</i>	Slope	SE
RMSSD	Stroop test	8	-0.064	0.004	.880	-0.0008	0.0049
RMSSD	Cooldown	8	-0.469	0.220	.241	-0.0084	0.0064
SDNN	Stroop test	8	-0.299	0.090	.472	-0.0134	0.0175
SDNN	Cooldown	8	-0.426	0.182	.292	-0.0083	0.0072
pNN50	Stroop test	8	-0.385	0.148	.347	-0.0029	0.0029
pNN50	Cooldown	8	-0.388	0.150	.343	-0.0017	0.0017

Note. HRV changes were calculated as relative percentage changes from baseline. The slope represents the predicted change in self-reported stress per one-percentage-point increase in the respective HRV feature.

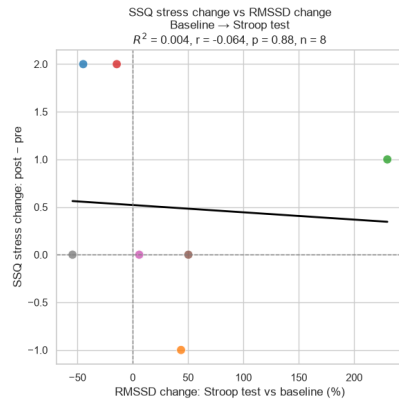
SE denotes the standard error of the slope. All *p*-values are unadjusted.

The corresponding recording-level scatterplots and fitted regression lines are shown in Figure 6.13. Each point represents one of Participant 1's eight recordings.

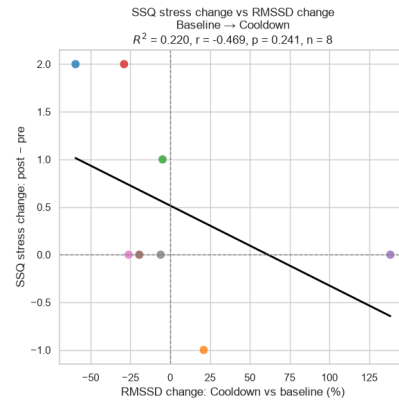
All six associations for Participant 1 were negative. As visible in Figure 6.13, however, the observations were widely dispersed, and several recordings exhibited comparatively large relative HRV changes. The fitted regression lines must therefore be interpreted cautiously given the small number of observations.

The weakest association was observed between the baseline-to- "Stroop test" change in RMSSD and self-reported stress change ($r = -0.064$, $R^2 = .004$, $p = .880$). The largest observed association was found for the baseline-to-"Cooldown" change in RMSSD ($r = -0.469$, $R^2 = .220$, $p = .241$), followed by the corresponding change in SDNN ($r = -0.426$, $R^2 = .182$, $p = .292$).

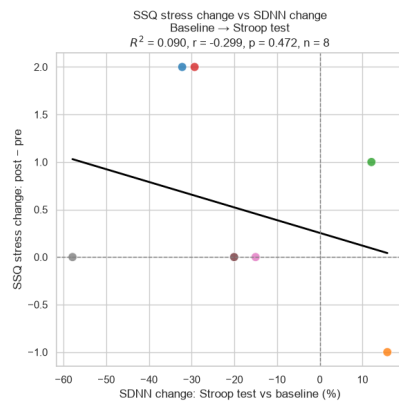
None of the six models had an unadjusted *p*-value below .05, and the R^2 values ranged from .004 to .220. Although all estimated associations were negative, the analyses were based on only eight repeated observations from one participant. They therefore do not



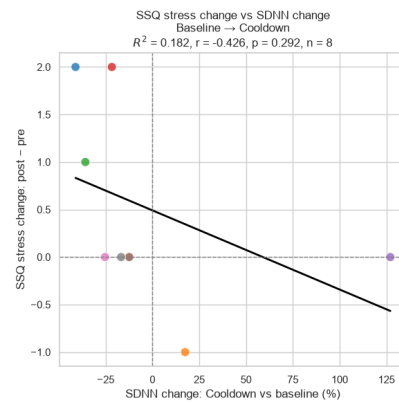
(a) RMSSD: Baseline to Stroop test



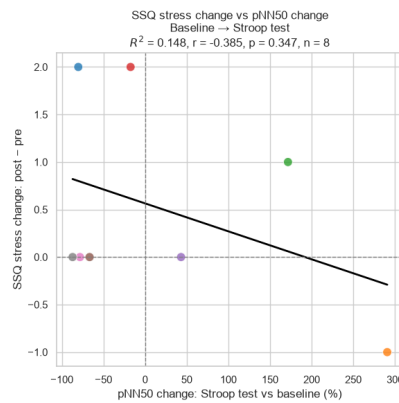
(b) RMSSD: Baseline to Cooldown



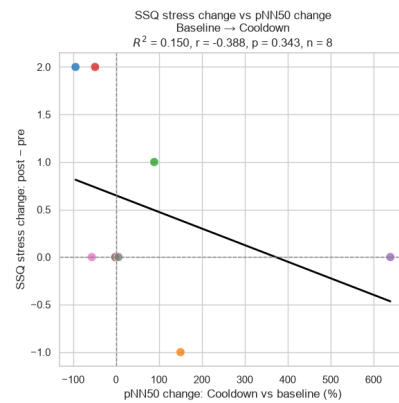
(c) SDNN: Baseline to Stroop test



(d) SDNN: Baseline to Cooldown



(e) pNN50: Baseline to Stroop test



(f) pNN50: Baseline to Cooldown

Figure 6.13. Associations between baseline-relative HRV changes and changes in self-reported stress across Participant 1's eight take-home recordings. The left column shows changes from "Baseline" to "Stroop test", whereas the right column shows changes from "Baseline" to "Cooldown". Each point represents one recording, the black lines represent fitted linear regression lines, and the dashed lines indicate zero change. The analyses are exploratory because they are based on only eight observations from one participant.

provide reliable evidence of a linear relationship between baseline-relative HRV changes and changes in self-reported stress.

Regression models were not fitted for Participant 2 because only three recordings contained the ECG data required for the baseline-relative comparisons. Furthermore, relative pNN50 change was available for only two of these recordings because baseline pNN50 was zero in recording 3. Participant 2's available observations were therefore examined only descriptively.

6.2.7 Summary of take-home feasibility

Both participants independently operated the system and created multimodal recordings without requesting assistance. All 14 recorded sessions contained usable head-movement data, numerical stress ratings, and verbal responses. However, only 11 sessions fulfilled the completeness criteria, as raw ECG was unavailable or insufficient in three recordings from Participant 2.

The take-home evaluation additionally documented one positional-tracking failure. The affected session was nevertheless completed and saved, and the participant independently repeated the recording. The results therefore demonstrate that independent repeated data collection was feasible for both participants, while also showing that the availability of raw ECG data was less consistent than that of the other recorded modalities.

7 Discussion

This chapter interprets the results with a focus on the development of the data-collection library and the associated stress-measurement protocol. The evaluation primarily examined whether the system could support supervised and unsupervised multimodal data collection in VR. The analyses of HRV and head movement were exploratory and do not establish the reliability or validity of individual stress biomarkers.

7.1 Interpretation of the Main Findings

The on-site evaluation indicates that participants were able to complete the measurement protocol using the developed application and library. The mean SUS score of 85, together with the absence of reported problems concerning the user interface or Polar H10 connection, suggests high perceived usability. The comparatively low score for the statement that participants would like to use the system frequently should not necessarily be interpreted as a usability problem. The application intentionally exposes participants to a stressor and is designed as a research instrument rather than for frequent voluntary use.

The Stroop Room also provided a suitable setting in which to evaluate the measurement protocol. All on-site participants reported higher stress after the test, with the mean rating increasing from 1.29 to 3.07. Verbal responses frequently described confusion and difficulties caused by the Stroop effect. These results support the assumption that the procedure generally increased perceived stress. However, individual responses also indicate that the effect was influenced by task strategy and perception. One participant reduced the difficulty by memorizing the room layout, while three participants had difficulty differentiating between yellow, orange, and red. Future implementations should improve the distinguishability of these colors without removing the intended Stroop interference.

The take-home evaluation provides initial evidence that the system can support independent data collection. Both participants operated the equipment without requesting assistance, and all recorded sessions contained head-movement data, numerical stress ratings, and verbal responses. One participant also recognized a positional-tracking failure, documented it in the verbal response, and repeated the session independently. The library and protocol therefore enabled repeated multimodal recordings outside a supervised laboratory

setting.

Nevertheless, only 11 of the 14 recorded sessions met the completeness criteria because raw ECG was missing from three sessions. Heart-rate data remained available, but the absence of raw ECG prevented the intended HRV analysis. The precise cause could not be identified. One possible explanation is insufficient signal quality, resulting in the Polar H10 not providing raw ECG data. Regardless of the cause, these results show that a successful sensor connection and the availability of heart-rate values do not guarantee that all required modalities are being recorded. This is an important weakness for unsupervised use and should be addressed in future iterations.

7.2 Interpretation of Potential Stress-Related Measures

The descriptive HRV results were partly consistent with the expectation that stress would be accompanied by reduced HRV. At group level, the mean RMSSD, SDNN, and pNN50 values were lower during cooldown than during baseline. Furthermore, all three features reached their lowest mean during the post-question phase or remained below baseline after the Stroop test. This could indicate that the physiological response continued after the task rather than corresponding exactly to the predefined application phases.

However, no consistent phase-related pattern was observed across participants. RMSSD and SDNN increased during the Stroop test after decreasing during the introduction, while individual trajectories differed substantially. Large standard deviations, skewed distributions, and crossing trajectories show that the group means did not represent a uniform physiological response. The take-home results showed similar variability between recordings from the same participant. These findings emphasize that HRV is highly individual and that a single group-level baseline may not be sufficient for interpreting a participant's current stress.

The regression analyses likewise found only weak or negligible associations between HRV changes and changes in self-reported stress, and none reached statistical significance. The absence of statistical significance should not be interpreted as evidence that no relationship exists. The on-site analyses were based on only 13 valid ECG recordings, while the take-home regressions used eight observations from one participant. In addition, the five-point stress scale provided only a coarse measure of subjective change. The sample size was appropriate for evaluating the prototype within the scope of this thesis, but insufficient for determining the reliability, precision, or validity of individual HRV features.

Consequently, RMSSD, SDNN, and pNN50 remain candidates supported by the reviewed literature, but the present results do not justify selecting one as a reliable standalone stress indicator. Their use should remain combined with individual baselines and subjective in-

formation. Larger studies with repeated measurements would be required to separate within-participant changes from general differences between participants.

The head-movement classification results provide only weak evidence that the recorded parameters can distinguish experimental phases. Logistic regression achieved a test balanced accuracy of .643 and correctly identified only half of the cooldown samples. The substantial overlap between phases in the UMAP and t-SNE projections further indicates that baseline and cooldown movement were not clearly separable.

Angular velocity trajectories had the largest absolute coefficients in the logistic regression model. This could be compatible with the initial assumption that sudden or reduced head movements may relate to stress. Nevertheless, coefficient magnitude alone does not demonstrate that these features are stress biomarkers, particularly given the limited classification performance. The target labels represented baseline and cooldown phases rather than independently verified stress states. Differences between these phases could also reflect posture, fatigue, adaptation, or position within the calibrated play space. The analysis therefore does not identify a movement parameter that can reliably determine stress. Instead, it demonstrates that head movement can be included in the collection protocol without an additional sensor and provides a basis for more controlled future investigation.

7.3 Implications for the Measurement Protocol

The results support a protocol that combines a standardized resting phase, physiological and movement recording, and subjective contextual information. The 90-second baseline represented a practical compromise between recording length and participant acceptance. Baseline and cooldown should use the same calming environment and instructions, with participants asked to relax and breathe normally. The observed individual and day-to-day variation indicates that repeated participant-specific baselines are preferable to relying only on general population values.

The sensor connection should remain explicitly participant-initiated. Participants should press a button to initiate the Polar H10 connection and receive clear confirmation when it succeeds. Displaying live heart rate is useful because it provides immediate feedback that the sensor is active. However, the take-home results show that this feedback should distinguish between the availability of heart-rate and raw ECG data. A future version should perform a signal check before starting the baseline, continuously verify that raw ECG samples are being received, and display a clear warning if the required signal becomes unavailable. This would reduce the risk of completing an apparently successful session that cannot subsequently be used for HRV analysis.

Combining numerical stress ratings with verbal responses also proved useful. All partici-

participants successfully recorded verbal feedback, and the take-home responses documented daily circumstances and technical problems that would otherwise have remained unknown. Informally, participants appeared willing to share substantial contextual information through the recording. Such responses can therefore complement standardized ratings, although they should not replace structured measures.

The lack of reported stress changes in the later take-home sessions suggests that repeated exposure may alter the measurement procedure. Habituation, learning, changes in strategy, or the limited sensitivity of the five-point scale are possible explanations, but cannot be distinguished with two participants. Repeated-measures protocols should therefore consider adaptation when selecting a stressor and interpreting participant baselines.

Overall, the main contribution of the evaluation is the implementation of a usable protocol that integrates ECG, heart rate, head movement, numerical ratings, and verbal context. Participants were able to operate the system independently, supporting the feasibility of multimodal VR data collection. However, raw-ECG completeness must be monitored more reliably before the system can be used for larger unsupervised studies. The physiological and movement analyses illustrate how the collected data can be examined, but they do not validate any individual parameter as a stress biomarker.

8 Conclusion

8.1 Summary of Contribution

This thesis investigated the feasibility of collecting and storing multimodal data during stress-related virtual reality interventions. The work addressed the practical challenge of conducting repeated measurements outside a laboratory, where participants should be able to operate the required equipment independently and where the recorded data should remain available for subsequent scientific analysis. The thesis therefore focused primarily on the development and evaluation of a data-collection system and protocol, and not on the clinical validation of individual stress biomarkers.

The main technical contribution is a reusable data-collection library for Unity applications running on a standalone Android-based VR headset. The library provides a common structure for recording and storing different data modalities and can be extended by implementing additional recorder components. In the application developed for this thesis, the library was used to collect heart-rate and raw ECG data from a Polar H10 heart-rate monitor, high-frequency head-movement data from the Meta Quest 3, self-reported stress ratings, application events, task-performance data, and voice recordings. A custom Kotlin-based Android library was developed to support Bluetooth communication with heart-rate monitors and ECG streaming from the Polar H10.

A VR application containing a virtual Stroop Room was developed to evaluate the library in a stress-related experimental procedure. The application combined a 90-second baseline phase, a Stroop Room stressor, subjective stress ratings, verbal responses, and a cooldown phase. Data were stored on the headset using timestamps, phase information, and pseudonymous identifiers. The resulting recording archives could subsequently be transferred to encrypted storage for further processing.

The system was evaluated in two complementary settings. The supervised on-site evaluation included 14 participants, while the exploratory take-home evaluation included two participants who were asked to use the system once per day for seven days. The on-site evaluation demonstrated good perceived usability. SUS scores ranged from 75 to 100, with a mean score of 85. All participants completed the application and provided verbal feedback. In addition, all participants reported higher perceived stress after the Stroop Room

test. The mean stress rating increased from 1.29 before the test to 3.07 after the test, suggesting that the procedure generally produced the intended subjective stress response.

The take-home evaluation showed that the system could be used independently for repeated multimodal recordings. Both participants operated the equipment without requesting technical assistance. All recorded sessions contained head-movement data, numerical stress ratings, and verbal responses. However, raw ECG data were missing or insufficient in three sessions, meaning that only 11 of the 14 recorded sessions fulfilled all completeness criteria. The results therefore demonstrate the feasibility of independent multimodal recording while also identifying raw ECG availability as an important weakness of the current implementation.

8.2 Relation to Research Questions

The main research question asked to what extent a standalone VR application and reusable data-collection library can support the collection and storage of multimodal data in supervised and home-based settings. The results indicate that the developed system can support this process to a considerable extent. The on-site evaluation showed that participants were able to complete the application and interact with the recording procedure. The take-home evaluation further showed that both participants could independently operate the system and produce repeated recordings without direct researcher assistance. The library successfully integrated data from the Polar H10 and Meta Quest 3 with subjective ratings, application events, and audio responses. The use of timestamped files, phase labels, pseudonymous identifiers, and local storage provided a structured basis for subsequent data management and analysis.

However, the results also show that successful operation of the application does not necessarily guarantee complete availability of all recorded modalities. Heart-rate data were available during the take-home sessions, but raw ECG data were missing or incomplete in three sessions. Consequently, the system can be considered feasible as a prototype for multimodal data collection, but additional quality-control and error-detection mechanisms are required before it can be used reliably in a larger unsupervised study.

The first subquestion concerned the variation of HRV features across the phases of the VR protocol and their association with self-reported stress. RMSSD, SDNN, and pNN50 were calculated from ECG recordings for the five application phases. At the group level, the three features generally showed lower values during cooldown than during baseline. Nevertheless, the phase-related trajectories varied considerably between participants. The group means were affected by large standard deviations, skewed distributions, and substantial differences between individual trajectories. The take-home recordings showed similar vari-

ation between recording days from the same participant.

The regression analyses did not provide evidence for a clear linear association between baseline-relative changes in RMSSD, SDNN, or pNN50 and changes in self-reported stress. None of the six on-site models reached statistical significance, and the explained variance was small. The participant-specific take-home analyses were similarly inconclusive. Thus, the results do not support selecting one of the investigated HRV features as a reliable standalone indicator of stress. Instead, RMSSD, SDNN, and pNN50 should be regarded as suitable candidate features for exploratory stress research. Their interpretation should take individual baselines, repeated measurements, and subjective or contextual information into account.

The second subquestion concerned the movement parameters tracked by the Meta Quest 3, their storage for stress-related analysis, and their ability to distinguish between the baseline and cooldown phases. The application recorded positional and rotational trajectories as well as linear and angular velocity data at high frequency. The data were stored together with timestamps and phase labels, allowing them to be separated and processed after the recording sessions.

The classification analysis provided preliminary evidence that the recorded head-movement data contained information related to the experimental phases. Among the evaluated models, the k -nearest-neighbours classifier achieved the best performance, with a test accuracy of 0.783 and a balanced accuracy of 0.732. Position trajectories, particularly PosX and PosZ, were most influential for this classifier, while angular velocity trajectories had the largest absolute coefficients in the logistic regression model. These results show that head movement can be captured and used for exploratory analysis without requiring an additional sensor.

Nevertheless, the classification labels represented baseline and cooldown phases rather than independently verified stress states. Consequently, the classification results do not demonstrate that any individual movement parameter determines stress. Differences between the phases may also have resulted from posture, fatigue, familiarity with the application, or the participants' position within the virtual environment. Head movement should therefore be regarded as a potentially useful complementary modality for stress-related studies rather than as a validated standalone stress measure.

The third subquestion concerned the completeness and usability of repeated multimodal recordings collected independently at home. Both take-home participants completed recordings without requesting assistance. Across the 14 recorded sessions, 11 were complete and three were partial. All sessions contained usable head-movement data, numerical stress ratings, and verbal responses. The recordings also demonstrated that audio responses could provide contextual information about daily events, technical problems, and

factors that participants perceived as relevant to their stress. These results support the usability of the protocol for exploratory home-based recording. At the same time, the incomplete ECG recordings show that usability and data completeness are separate requirements. The protocol was therefore operationally feasible, but not yet sufficiently robust to ensure complete physiological recordings in every unsupervised session.

8.3 Limitations

The findings must be interpreted in light of several limitations. First, the study used small convenience samples. The on-site evaluation included 14 participants and the take-home evaluation included only two participants. The participants were recruited from one institution and did not represent a clinical target population. The findings consequently provide evidence of technical feasibility and usability, but they cannot be generalized to patients, nurses, or other intended user groups.

Second, the study did not formally validate the recorded biomarkers. No clinical-grade ECG reference system, randomized control group, or independent ground-truth measure of stress was included. The subjective stress measure consisted of a single five-point question and therefore provided only a coarse assessment of perceived stress. Although all on-site participants reported increased stress after the Stroop Room test, the task did not constitute an independently validated stress induction for every participant.

Third, the study did not perform formal test–retest or intra-subject reliability analyses. The repeated take-home recordings provided an initial view of day-to-day variation, but the number of participants and the uncontrolled recording conditions were insufficient for reliable statistical estimation. Time of day, sleep, exercise, caffeine consumption, food intake, medication, posture, and environmental conditions were not standardized. These factors may have contributed to the observed variation in HRV and head movement.

The availability of raw ECG data was another important limitation. The application indicated that heart-rate data were available, but this did not guarantee that raw ECG data required for HRV analysis were being recorded. The causes of the missing ECG data could not always be reconstructed from the stored files. The head-movement analysis was also limited by the small dataset and by the use of baseline and cooldown as indirect proxies for stress. Finally, the verbal responses were transcribed and summarized by one researcher without formal qualitative coding or independent verification. They are therefore useful as contextual information, but do not constitute a formal qualitative analysis.

8.4 Future Work

Future work should first improve data-quality monitoring during recording. Before a session begins, the application should verify not only the availability of heart-rate values but also the reception and quality of raw ECG data. This verification should continue throughout the session, and participants should receive a clear warning if a required modality becomes unavailable. Automatic completeness checks, more detailed diagnostic logs, and recovery mechanisms for interrupted recordings would further improve the reliability of the system. On-device encryption and an encrypted upload mechanism could also be investigated to reduce the need for manual data transfer while maintaining appropriate data protection.

The system should next be evaluated in a larger and more diverse longitudinal study. A recording period of several weeks or months would allow researchers to investigate within-participant baseline stability, day-to-day variation, and habituation to the Stroop task. Relevant contextual factors, such as sleep, exercise, caffeine consumption, medication, posture, and time of day, should either be standardized or recorded systematically. Future studies should also compare Polar H10 recordings with a reference ECG system and formally evaluate signal quality, test–retest reliability, and intra-subject reliability.

Further head-movement research should use participant-independent validation, such as leave-one-participant-out cross-validation, to reduce the possibility that models learn participant-specific movement patterns. Additional movement representations and features could be investigated, while the effects of posture, task interaction, and virtual-environment layout should be controlled. A multimodal analysis combining HRV, heart rate, head movement, voice, self-report, and task performance may be more informative than analyses based on a single modality. Because the results showed substantial individual variation, personalized baselines and participant-specific models may be more appropriate than general population thresholds.

In conclusion, this thesis demonstrates that a standalone VR application and reusable data-collection library can support supervised and independent home-based collection of multimodal data relevant to stress research. The developed system provides a practical foundation for future XR resilience studies and demonstrates that head movement, physiological data, subjective ratings, and verbal context can be recorded within one application. The results do not yet validate individual HRV or movement parameters as stress biomarkers. They instead establish technical feasibility, identify the requirements for reliable unsupervised recording, and provide a basis for larger studies investigating personalized and multimodal stress responses.

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Appendices

A Experiment Protocol and Informed Consent Form

Test Protocol (estimated duration 30 minutes)

Biomarker Recording in Stroop Room

This document is intended to be a protocol and guideline on how to conduct testing to evaluate the biomarker recording module developed for XR2esilience.

Measures. heart rate, ECG, head movement, two single question questionnaires about perceived stress state, one short audio recording after stress intervention, age, gender, system usability score questionnaire

Roles and Materials

Hardware: Meta Quest 3 with application installed; Polar H10 heart rate monitor + chest strap

Documents: participant information sheet; informed consent form

Utilities: Alcohol wipes; water; salted crackers

Briefing

“Thank you for taking part in this experiment. Today you will try a short VR intervention in which we measure your heart rate data with a Polar chest strap and record your head movements. I am evaluating the recording system, not your personal health. That means I will record your biometrics and evaluate the data in aspects of collection quality. I will and cannot provide any medical guidance based on that data.

If you feel uncomfortable or nauseous at any time, please tell me and we will pause or stop the session. You can withdraw from the study at any point without consequences.”

(obtain written informed consent)

Setup

“First, I want to ask you for some biographic data. Please fill out this form with you age and gender”

(fill out demographics form)

“During the experiment, the application will go through four phases. First, it will lead you through the connection setup with the Polar belt. Second, there is a baseline recording phase of 90 seconds followed by a quick question about perceived stress. Then there

will be the Stroop Room task, a mild stressor that will get you moving a bit. After the Stroop task, you will be asked to answer the perceived stress question again and to describe your experience with the application verbally. Lastly, there will be a 90 second cooldown period after which the application will close automatically. After the VR session I will ask you to fill out a short usability questionnaire as well. Do you have any questions?”

(time to answer questions)

“I will now give you a Polar HRM. Please put it on that device is placed right below your chest in the center. Make sure the device has a snug fit, but it does not hurt you. You may adjust the strap length.”

(give participant the HRM and privacy to put it on)

“Next, I will give you the VR headset. Let me briefly explain how to put it on and how the controller's work.”

(adapt explanation detail according to participant's experience with VR)

“Please follow the instructions inside the headset, you should start in a green meadow with an UI in front of you. Make sure that after putting on the headset you recenter once by long pressing the Meta button on the right controller. The application will close automatically once the experiment is over.”

Recording

Let the participants go through the intervention. If the participant asks for help with the Interface, we can help them by providing hints.

Observe the participant and intervene if you notice any safety or comfort issues.

Post task questionnaire

“Your task in Virtual Reality is now over. Do you feel nauseous in any way? May I offer a glass of water?”

(give participant time after the VR task to recover shortly)

“I will now ask you to fill out this System Usability Scale (SUS) questionnaire, please take your time.”

(give participant SUS questionnaire)

Debrief

“Thank you very much for participating. The experiment is over now. Do you have any questions for me?”

Data collection

Connect Meta Quest 3 to computer with a USB cable and retrieve the recorded zip archive. The zip archive is stored under the `Android/data/at.ustp.biomarkers/files` path. The filename will be in the format “[user_id]_[unix_timestamp].zip”.

Inside the archive there will be the following files:

- [user_id]_[unix_timestamp]_voice.wav
- [user_id]_[unix_timestamp]_[uuid]_HR.csv
- [user_id]_[unix_timestamp]_[uuid]_HEAD.csv
- [user_id]_[unix_timestamp]_[uuid]_ECG.csv
- [user_id]_[uuid]_questionResponses.json

Experiment Timing Template

- Briefing (3-5 min)
- Setup (5-8 min)
- Recording (6-10 min)
- Post task questionnaire (4-6 min)
- Debrief (2-3 min)
- Data collection (2 min)

Informed Consent Form

Researcher: Georg Becker

Email: it241503@ustp-students.at

You are invited to take part in a research study evaluating a virtual reality biomarker recording module developed for XR2esilience. Before you decide whether to participate, please read the information below carefully. Ask any questions you may have before signing this form.

Your participation is voluntary.

The purpose of this study is to evaluate the technical performance and usability of a virtual reality recording system. The study focuses on the quality and feasibility of collecting biometric and interaction data during a short VR task.

This study is not a medical examination and is not intended to assess, diagnose, or treat any health condition. No medical advice will be given based on the recorded data.

If you agree to participate, you will be asked to:

- provide basic demographic information:
 - age
 - gender
- wear a Polar H10 heart rate monitor with chest strap
- wear a Meta Quest 3 VR headset
- complete a short VR session lasting about 30 minutes in total
- complete:
 - a 90-second baseline recording
 - a brief perceived stress question
 - the Stroop Room task, which is intended as a mild stress task and may involve some movement
 - a second brief perceived stress question
 - a short verbal description of your experience, which will be audio recorded
 - a 90-second cooldown period
- complete a short System Usability Scale (SUS) questionnaire after the VR session

The researcher may help if you need help operating the interface.

During the study, the following data may be collected:

- heart rate data
- ECG-related data from the Polar H10 device
- head movement data recorded through the VR system
- answers to two single-item perceived stress questions
- a short audio recording of your verbal feedback after the stress task
- age
- gender
- usability questionnaire responses

Your participation will take approximately 30 minutes.

The study involves minimal risk, but possible discomforts include:

- mild stress or frustration during the Stroop Room task
- temporary discomfort from wearing the chest strap
- VR-related discomfort such as dizziness, nausea, eye strain, disorientation, or fatigue
- minor inconvenience from being audio recorded

If you feel unwell, uncomfortable, or wish to stop at any time, please tell the researcher immediately. The session can be paused or ended at once.

You may not receive any direct personal benefit from participating. Your participation may help improve the design and evaluation of VR-based biomarker recording systems.

Your participation is entirely voluntary.

You may:

- choose not to participate
- refuse to answer any question
- stop the session at any time
- withdraw from the study without giving a reason
- withdraw without any negative consequences

If you withdraw, no further data will be collected. You may request deletion of identifiable data collected up to the time of withdrawal.

You will not receive compensation for participating.

Your data will be handled confidentially and in accordance with applicable data protection laws, including GDPR.

Data privacy is ensured during the preparation and execution of the study. Your data will be categorized as follows:

1. **Personal data** that directly identifies you (Name, Age, gender)
2. **Pseudonymized data**, where no identifying data is stored alongside the collected measurements

Measurement data will be stored in a **pseudonymized** format using a unique randomized key (UUID e.g. **83f2f0fb-8dbd-422b-9bb5-a634fbbfdbc3**) Biosignals will only be linked to this identifier.

To allow for demographic analysis and to fulfill GDPR rights of data subjects (Access, Correction, Erasure, ...) the UUID key will be stored alongside the demographic data.

- Your data will be stored securely and encrypted on the researcher's (Georg Becker) IT infrastructure.
- Your name will not appear in reports, publications, or presentations.
- Only authorized members of the research team will have access to identifiable data.
- Audio recordings will be used only for research purposes described in this study.

By signing below, I confirm that:

- I have read and understood the information in this form.
- I have had the opportunity to ask questions and received satisfactory answers.
- I understand that participation is voluntary.
- I understand that I may stop participating at any time without penalty or negative consequences.
- I understand what data will be collected and how it will be used.
- I understand that this study is not a medical examination and that no medical advice will be provided based on the collected data.
- I freely agree to participate in this study.

Participant

Name: _____

Age: _____

Gender: _____

Signature: _____

Date: _____

B Instructions on how to start application during week long evaluation

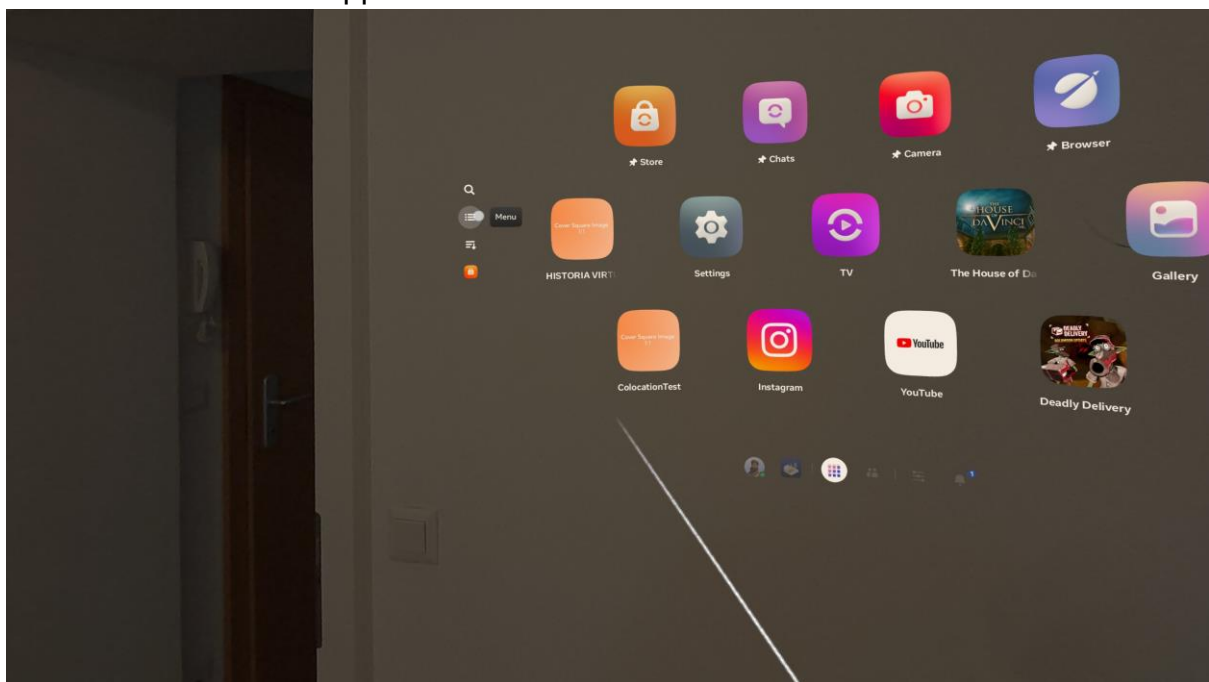
One weeklong recording of stress related biomarkers in VR

Thank you very much for taking part in the data collection for my master thesis. In this thesis I study how stress related biomarkers can be recorded and stored on VR headsets for use in take-home settings.

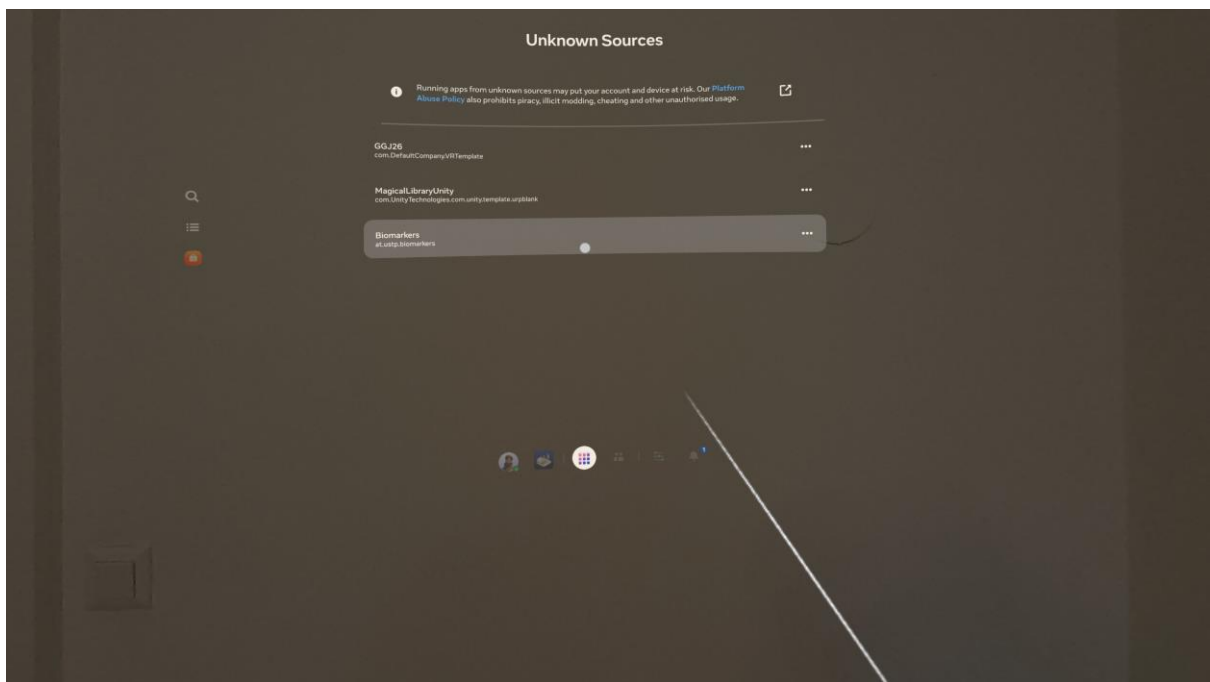
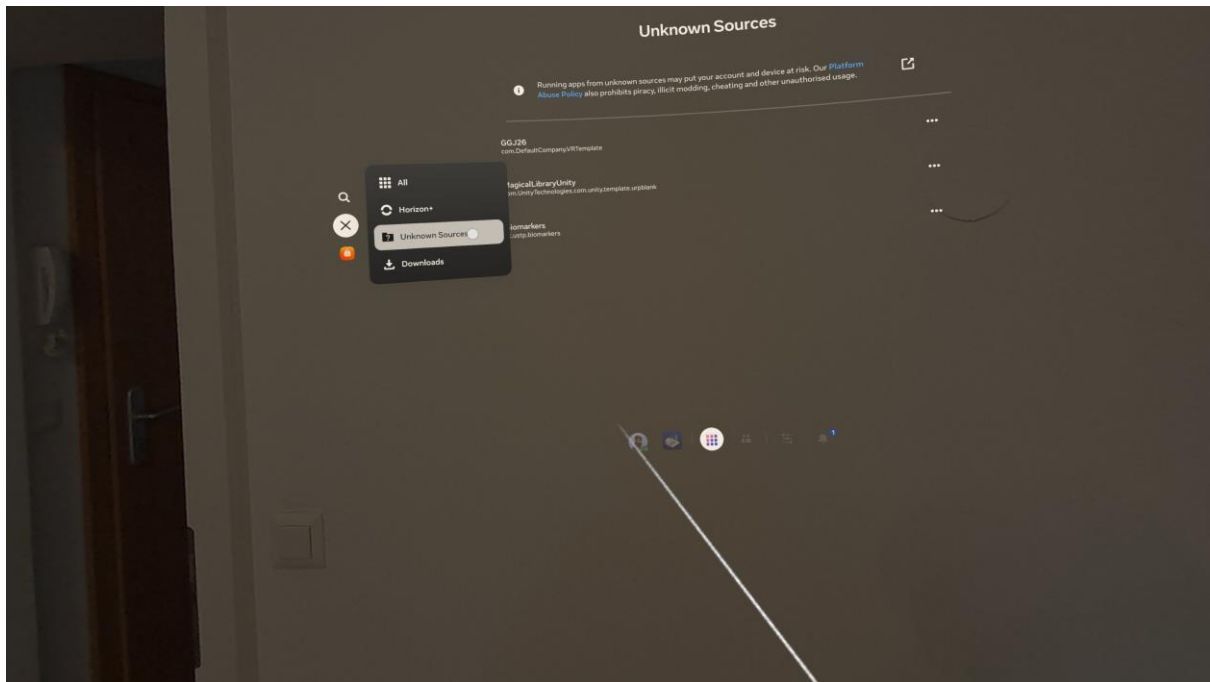
In this part of the study, you will get a Meta Quest 3 VR headset and a Polar H10 HRM device to take home for the span of a week. Once per day, please use the Biomarkers application on the VR headset while wearing the Polar H10 from start to finish.

In case you run into any difficulties during the study, please contact me via email (it241503@ustp-students.at) or contact me via telephone (+43 680 550 21 16).

To start the application, you need to navigate to the Unknown sources' menu and select the Biomarkers application.



To open this screen, press the Meta button on the right controller.



C Transcripts of on-site evaluation audio recordings

“ Also das Spiel ist sehr interessant und gut, ich find das gut umgesetzt. Ahh, manchmal ist es schwierig wenn man sich ganz umdrehen muss. Ahh, aber ich finde den Raum ansehnlich gestaltet und eigentlich ganz cool. ”

“ Soll ichs auf Englisch sagen, I guess, joa. ÄÄhm, Aso das Problem war eher, dass ich es nicht ganz direkt sofort verstanden habe. Mit dem Word und der Color, aber dann gings eigentlich. Es war eigentlich ganz lustig, ich kenn das Spiel schon, ja. ”

“ Also ich am Anfang, ähm, was quasi Color bedeutet aber ich glaub das liegt an mir weil ich die instructions nicht gut durchgelesen hab. Also weil ich das gecheckt hab das Gefühl es ging eigentlich ganz gut. Ich hatte zwischendurch immer wieder so Hänger, wo ich dann kurz nachdenken musste und es ein bisschen länger gedauert hat. Ähmm, aber ich würd sagen ich hatte sonst eigentlich keine großen Probleme, und es war jetzt nich überdurchschnittlich stressig für mich oder so. Es ähh, es hat gepasst. Ähmm, what parts I enjoyed particulary? Aso ich find das Layout ansich von dem Spiel eigentlich ganz nice. Was halt bisschen, ääähm, was ein bisschen unpraktisch manchmal war ist das halt so rot und gelb genau hinter mir sind. Da muss ich mich immer so nach hinten drehen. Und manchmal hab ich mich dann so in die falsche Richtung gedreht und dann hab ich mich so ein bisschen äähm verdrehen müssen aber. Ja war lustig, war lustig. ”

“ There were no problems I have encountered during the module. And I enjoyed how fast paced the game was. ”

“ So the process was, äähm, so it was difficult to answer the scale first because aahm, initianially ääh the scale of one to five which I would've put on four in the very beginning but over time it became much less stressful äähm, which is why gave it a two at the end of the test. I don't know how long the test was but I guess maybe two minutes. Aahm but äähm, at the during the test I got to know the layout of the room so that I only looked at the prompts and started ääh, clicking in the direction of the colors without even looking in them. Which made it less stressful, so I could focus on the prompt and yea sort of like getting used to it quite quickly and easily. Aaahm, anything else? I think struggled with mostly, äääh, getting the color right. I was able to get the words mostly right in the stroop test. Aaaaah, yea I think those are the things. Yap! ”

“ Soo, I was very confused with the rules, and I got very stressed and mixed up, aahm, the tasks even tho sometimes I got them right. ”

“ It was fun. I mixed up some of the colors and words. But in total I think very cool that you have to turn around in a circle. That it is not just a wall. You have to really think in what direction to turn. And I also liked there is not just one screen that I can see the colors but

there is many different directions. But it is a bit confusing to always have to circle around. But at some point you remember I remembered the colors where they are approximately and it was fine. ”

“ Ich hab nicht gleich checkt, dass, rund um mich herum die Bildschirme sind die mir das sagen. Das war dann leichter. Und ich hab einfach ständig vergessen, in welche Richtung welche Farben stehn. Aber das ist ein me Problem, kein Problem von dem Ding. ”

“ Aaaaaah, ich hatte Probleme mit dem Verständnis von dem Problem. Also mit dem Verständnis von der Aufgabe selbst. Ich glaub ich habs nicht ganz Verstanden die, vielleicht bin ich einfach nur stupid oder hatte einen langen Tage aber irgendwie hab ich nicht die Instruktionen so ganz mitbekommen. ”

“ Ähm, was ich schwierig gefunden hab war das die Farbe manchmal, also die Farbe vom Text manchmal ned die selbe Farbe war wie also scho ich weiß welches orange zu welchem orange gehört aber es ist nicht genau der gleiche shade. Äähm, genau und manchmal sind ein paar Tasks wo es sehr wo es sehr schnell geht und dann bleib ich auf einmal hängen weil ich glaub mein Gehirn nicht mehr registrieren kann. Äähm, was jetzt Color und Text und was ich eigentlich siech. Aber dann gehts eh wieder. ”

“ Also im Grunde aaah. Na was hab ich jetzt? Im Grunde also die Ganze Welt schaut okay aus. Aaah das aahh bissl verwirrend ist halt beim Stroop Room wenn die Farben daherkommen, dann die Geschwindigkeit durchgeht. Und ahhh, weiß nicht aber irgendwie die Farbe von ahh, vom Ladenbalken verwirrt mich paar also so ein bisschen. Ahhh, nachdem am anfang hab ich die Farbe gesehen. Ist dann zum Beispiel Orange gestanden in Grün hab aber dann irgendwie nicht gleich überrissen das drüber steht Word oder Color. Also das war jetzt so ein Problem bei mir halt. Aaah was ich am liebsten wollte war dann wo ich endlich dann halt drinnen war. Äähm jo also weiß jetzt gar nicht wie ich das sagen soll. Soll ich aufhören zum recorden? ”

“ Recording Also du willst jetzt ein dings haben, (mumbles) Lesekompetenz ist mein Hauptproblem aber ansonsten find ich es sehr witzig. Mann könnts Grafisch noch ein bisschen aufhübschen aber es passt gut. ”

“ So overall it was pretty clear but I feel like that when the word was aaahm yellow or orange the difference of the hue of them both is not that noticeable but overall yea it's a stressed test I guess I'm pretty stressed out. ”

“ Ähm probleme waren nur zu verstehen wann ich die Farbe und wann das Wort zählt und das ich rundherum alle Farben sehe. Und für mich war eine Schwierigkeit Rot und Orange aufeinander zu halten. ”

D Transcripts of week long at home evaluation audio recordings

D.1 Subject 1

Recording 1:

"Well, ahm, my overall stress level today was, I'd say a five out of ten, yeah. For sure!. I wouldn't say that I am particualary relaxed today. I still haven't finished my learning and my assignments so that's a bit off and what also impacted my stresslevel was that housekeeping told me that I am not allowed to use my, aahm, AC yeah that's it."

Recording 2:

"I am happy that I finished my task for today by, well the study related task for today. I am a little stressed because I still need to learn for my tommorows exam and I am also involved in the organisation of some things for the ÖH and things are not running all that smoothly but I am fine."

Recording 3:

" Aaahm so I had my exam today and I am having 2 more exams next week and yeah, failing stressed me out today."

Recording 4:

" I don't know wheter it was a bug or something but ususally when I turn my had I was able to move in the room but now my head staid fixed regardless where my head went and this was very distressing because I couldn't perform the challenges."

Recording 5:

" My, there were no particular stressors today, aahm I am looking forward for my work hours and to all the stuff I have to do today. "

Recording 6:

" Aahm, I am about to have an exam on Monday and on Wednesday and it's simply way to hot to learn and I am kinda worried if my avoidance of the heat would collide with my academic succes. "

Recording 7:

" Aah, tommorow I will have my first of two exams this week and I have learned a little yesterday and I will learn a little bit today and I hope that will suffice."

Recording 8:

“ I am looking forward to my test, and I am glad that I finished this 7 days study. ”

D.2 Subject 2

Recording 1:

“ Also heute war jetzt kein so besonders stressiger Tag, ich bin praktisch aufgestanden und hab IKEA Möbel gebaut was relativ wenig stresst, und dann bin ich essen gegangen also war mein Stresslevel auch recht low und dann hab ich wieder IKEA Möbel aufgebaut, also das war auch okay und jetzt bin ich hier und bin eigentlich noch immer noch nicht particulalry stressed weil es sind ja theoretisch Ferien.”

Recording 2:

“ Ja mein overall stress level heute war eigentlich mal ein up und auf und ab. Heut das erste mal seit Jahren wieder gechillt, das heißt am Anfang war ich gechillt, dann war ich gestresst und dann war ich wieder gechillt. Und jetzt stresst mich das ich so tief hier in der App drinstehe und das der Akku gleich leer ist. ”

Recording 3:

“ Ja heute war eigentlich Stressmäßig, ne fünf von zehn. Also ich bin, ich musste heut morgen relativ viele Sachen erledigen und hab nicht so gut geschlafen und Fitnessstudio lief nicht so gut, deswegen war ich sehr gestresst. Aber jetzt hab ich gerade 3 Stunden die Odyssee von Christopher Nolan angeguckt. Das hat mich sehr excited, es hat mich aber auch ein bisschen entspannt, weil ich hatte ein bisschen Sorge, dass der Film nicht gut ist und das die ganzen Kritiker recht haben aber ich bin sehr großer Nolan Fan deswegen hät mich das sehr belastet wenn das schlecht gewesen wäre. Aber es war gut und ich hatte nen gechillten Abend und das ist jetzt noch das letzte bevor ich mich mit einem Buch ins Bett lege.”

Recording 4:

“ Mein overall stresslevel today ist eigentlich recht moderat weil heut ein sehr entspannter Tag war aber wir haben morgen nen Dreh deswegen bin ich dann doch den ganzen Tag unterbewusst ein bisschen gestresst gewesen. ”

Recording 5:

“ Mein overall stresslevel ist eigentlich heute recht hoch, weil ich noch recht viel aufräumen muss heute und putzen muss und ich hatte heute actually nen ADHS test wo ich das gleiche Spiel spielen musste und davon brennt noch ein bisschen mein Kopf und deswegen

hat das Spiel mich wieder zurück zu meinem ADHS Test gebracht und jetzt fühl ich mich wieder gestresst. ”

Recording 6:

“ Mein Stresslevel war heute recht low weil ich heute eigentlich den ganzen Tag machen konnte was ich wollte und bisschen ÖH stuff machen konnte was sehr sehr satisfying war, das heißt ich war schon am Anfang vom Tag produktiv und somit war dann mein Stresslevel eigentlich Recht low und dadurch das ich den ganzen Tag Zeit hab und keinen Stress mit meinen Tasks hab bin ich eigentlich Recht entspannt. ”